Part 7
Dinosaurs!

7.0 Introduction and Recap

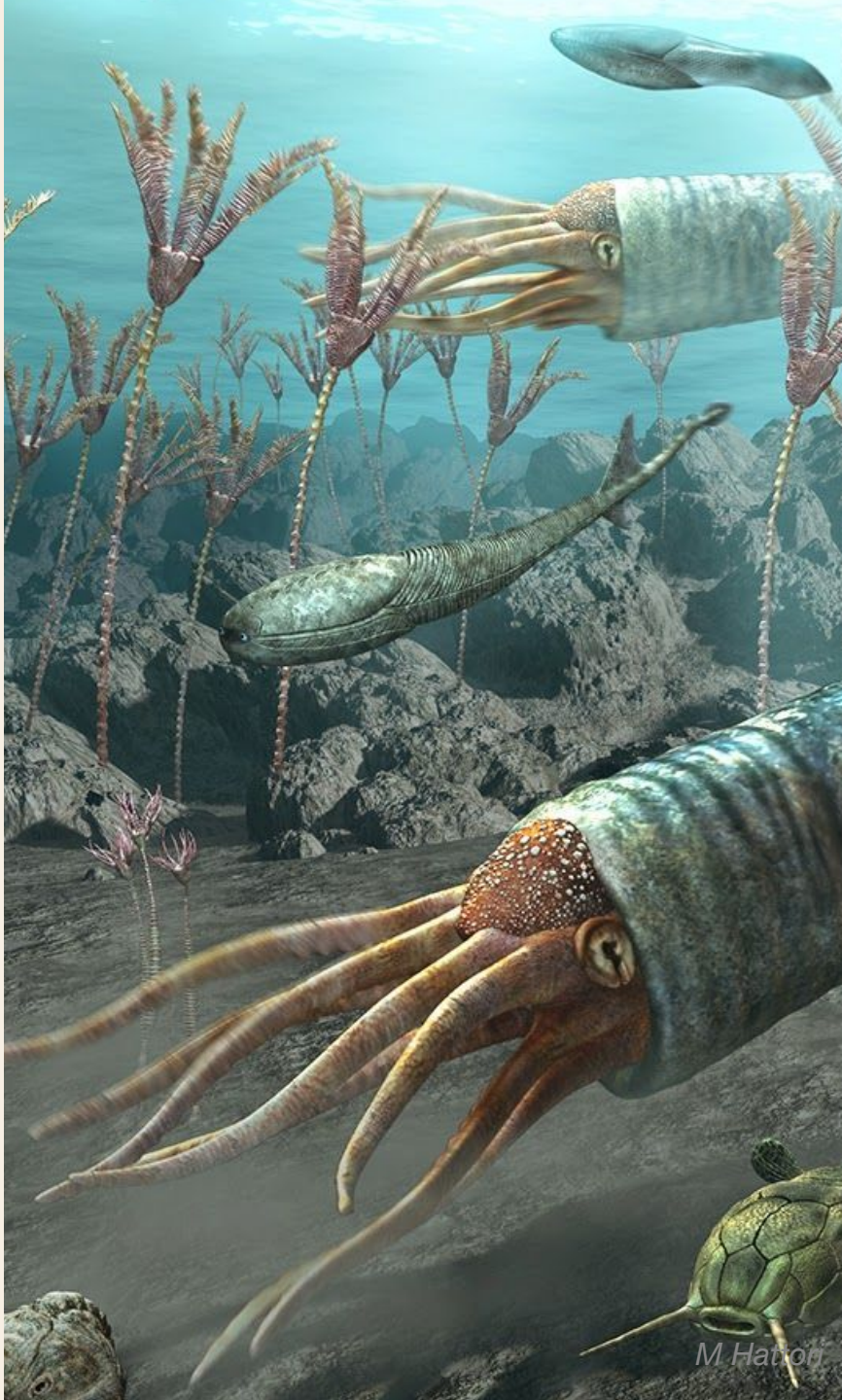
EART11100: How to Grow a Planet
Robert Sansom
University of Manchester

Cambrian Summary

- Appearance of lots of animal phyla
- Explosive diversification of unusual forms
- Particularly arthropods

Ordovician Summary

- Appearance of modern reef systems
- Great Ordovician biodiversification event
- Periods of cooling and warming related to extinctions



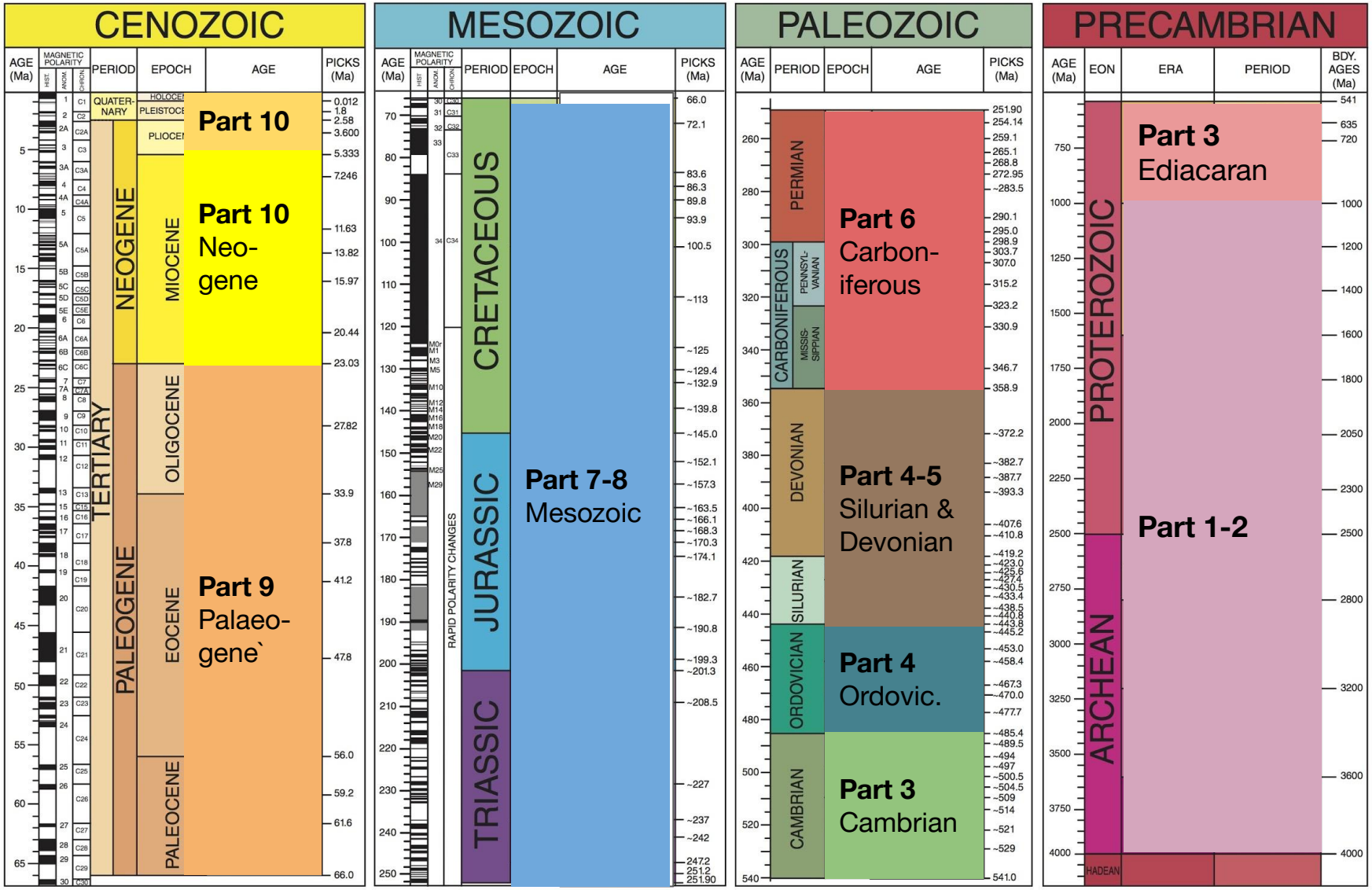
Silurian & Devonian Summary

- Diversification of vertebrates in the “Age of Fishes”
- Diverse jawless ostracoderms on the gnathostome stem-lineage
- Major groups of modern jawed vertebrates appear, including cartilaginous fish, ray finned fish, and lobe finned fish

Carboniferous Summary

- Arthropods and vertebrates have invaded the terrestrial realm
- Rainforests spread across the globe

GSA GEOLOGIC TIME SCALE v. 5.0



THE GEOLOGICAL SOCIETY OF AMERICA®

Walker, J.D., Geissman, J.W., Bowring, S.A., and Babcock, L.E., compilers, 2018, Geologic Time Scale v. 5.0: Geological Society of America, https://doi.org/10.1130/2018.CTS005R3C. ©2018 The Geological Society of America

*The Pleistocene is divided into four ages, but only two are shown here. What is shown as Calabrian is actually three ages—Calabrian from 1.80 to 0.781 Ma, Middle from 0.781 to 0.126 Ma, and Late from 0.126 to 0.0117 Ma.

The Cenozoic, Mesozoic, and Paleozoic are the Eras of the Phanerozoic Eon. Names of units and age boundaries usually follow the Gradstein et al. (2012), Cohen et al. (2012) , and Cohen et al. (2013, updated) compilations. Numerical age estimates and picks of boundaries usually follow the Cohen et al. (2013, updated) compilation. The numbered epochs and ages of the Cambrian are provisional. A "~" before a numerical age estimate typically indicates an associated error of ±0.4 to over 1.6 Ma.

REFERENCES CITED

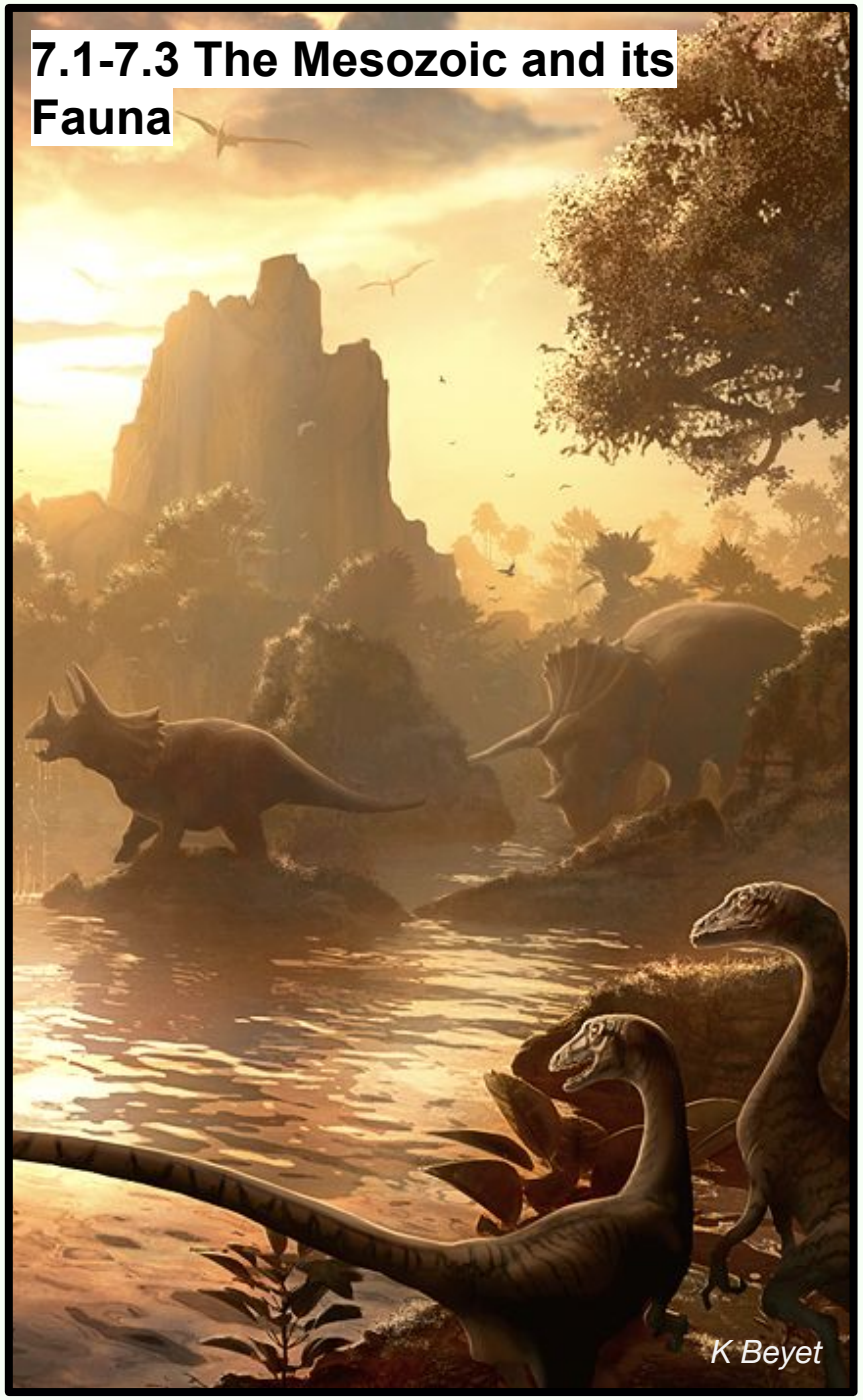
Cohen, K.M., Finney, S., and Gibbard, P.L., 2012, International Chronostratigraphic Chart: International Commission on Stratigraphy, www.stratigraphy.org (accessed May 2012). (Chart reproduced for the 34th International Geological Congress, Brisbane, Australia, 5–10 August 2012.)

Cohen, K.M., Finney, S.C., Gibbard, P.L., and Fan, J.-X., 2013, The ICS International Chronostratigraphic Chart: Episodes v. 36, no. 3, p. 199–204 (updated 2017, v. 2, http://www.stratigraphy.org/index.php/ics-chart-timescale; accessed May 2018).

Gradstein, F.M, Ogg, J.G., Schmitz, M.D., et al., 2012, The Geologic Time Scale 2012: Boston, USA, Elsevier, https://doi.org/10.1016/B978-0-444-59425-9.00004-4.

Previous versions of the time scale and previously published papers about the time scale and its evolution are posted to http://www.geosociety.org/timescale.

7.1-7.3 The Mesozoic and its Fauna



7.4-7.6 The “Jurassic Park”, reconstructing past life

Part 7
Dinosaurs!

7.1 Triassic

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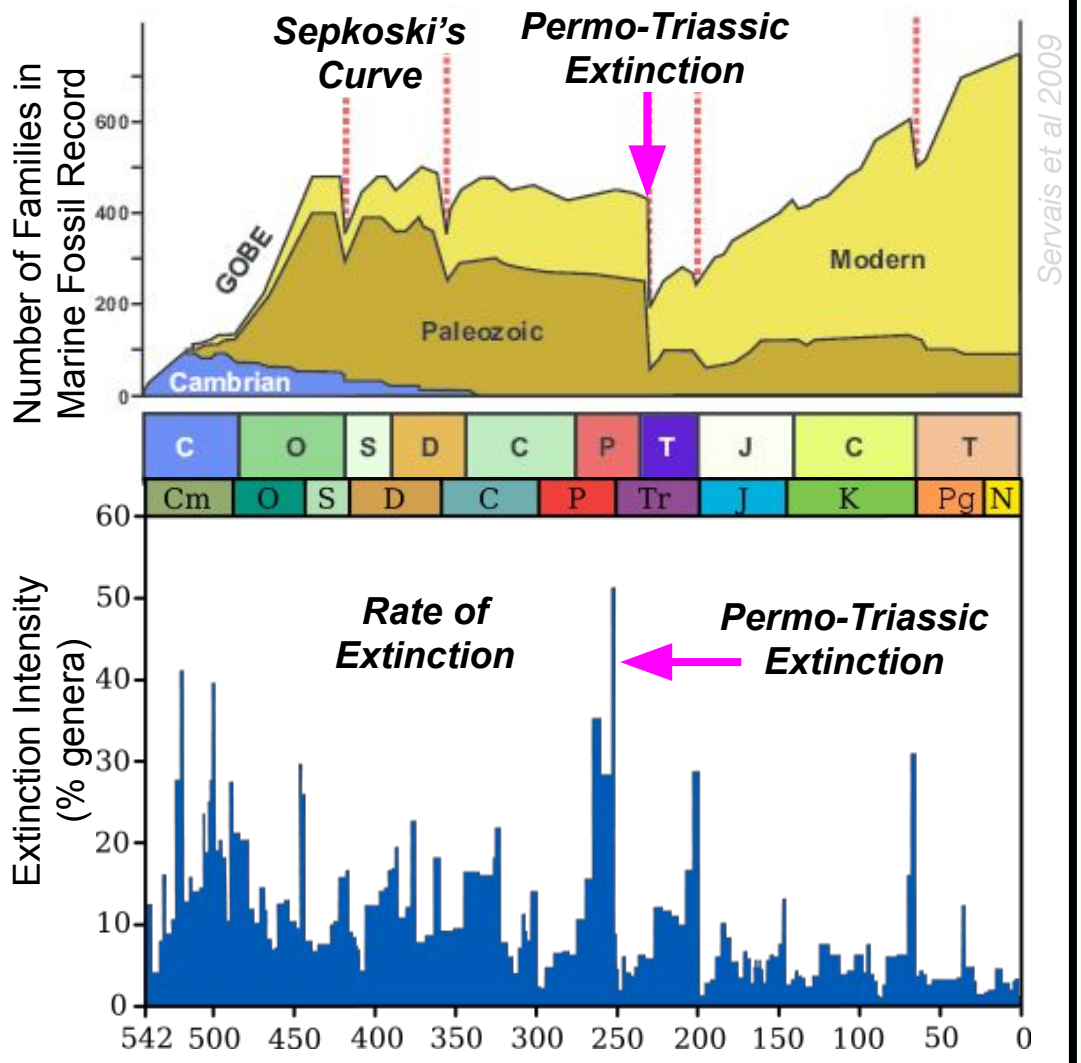
University of Manchester

When life nearly died

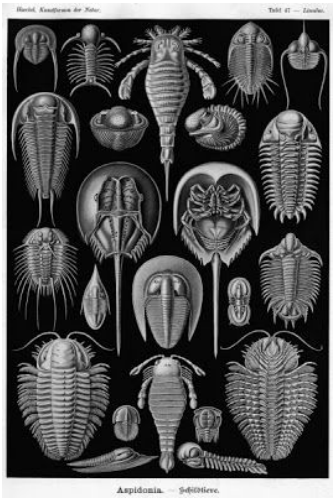


The biggest extinction event in the history of life on earth is the **Permian-Triassic Mass Extinction Event,** also known as "the Great Dying"

Many organisms affected: **83%** of all *genera* went extinct

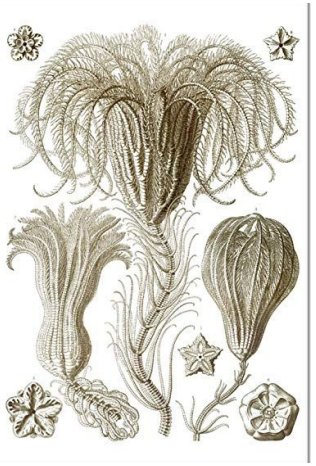
"Modern-type" faunas diversify afterwards in the Mesozoic and beyond, but "Palaeozoic-type" faunas never really recover

End Permian "Victims"



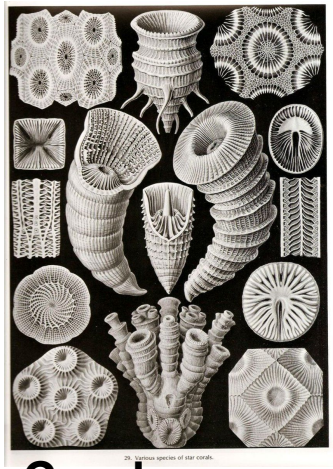
100% extinct

Trilobites



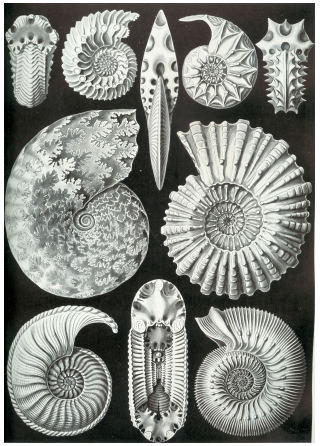
98% extinct

Crinoids



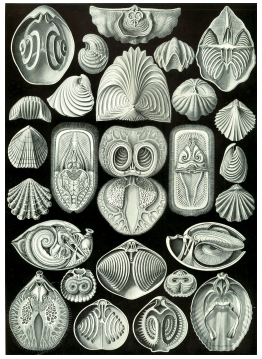
96% extinct

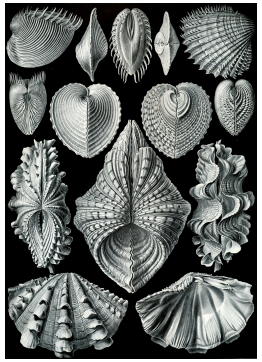
Corals

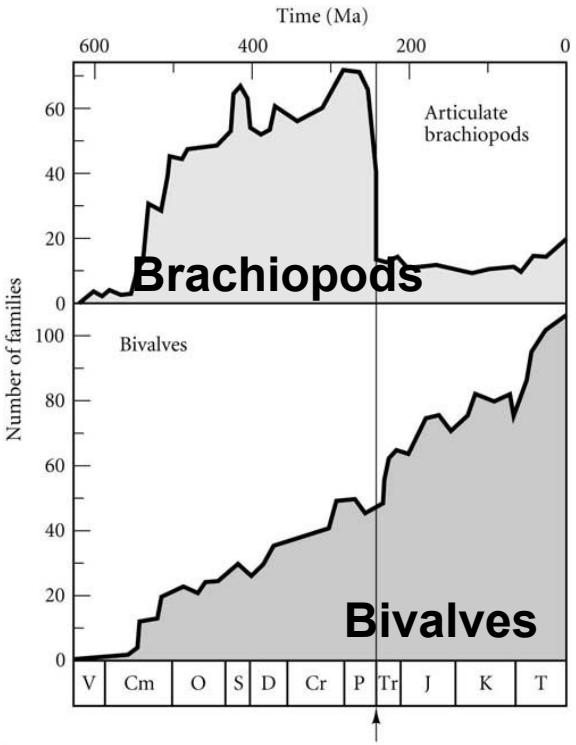


97% extinct

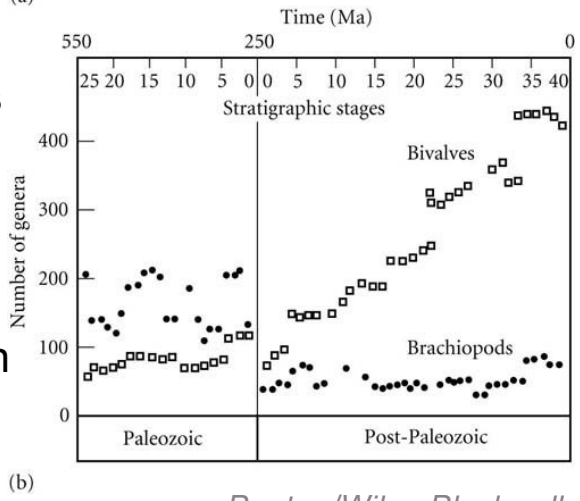
Cephalopods







Brachiopods (96% families extinct) replaced by **bivalves** (more modern shelled molluscs

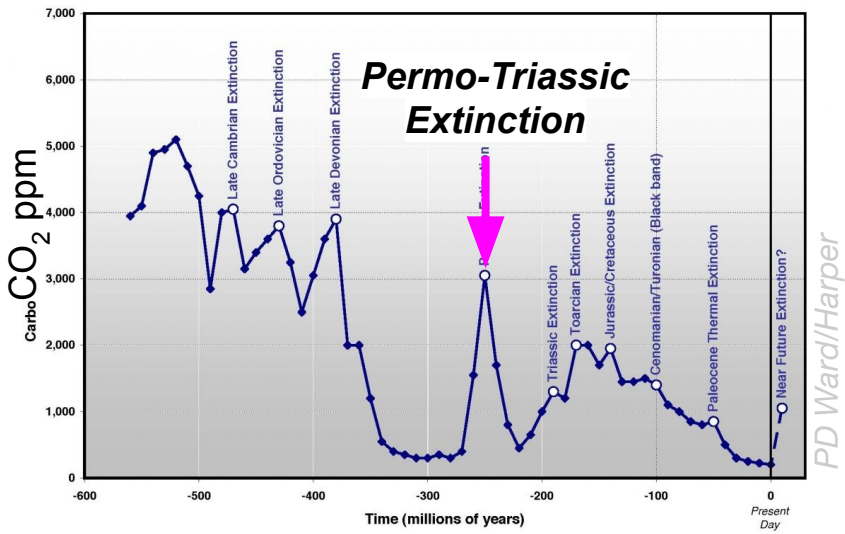


Benton/Wiley-Blackwell

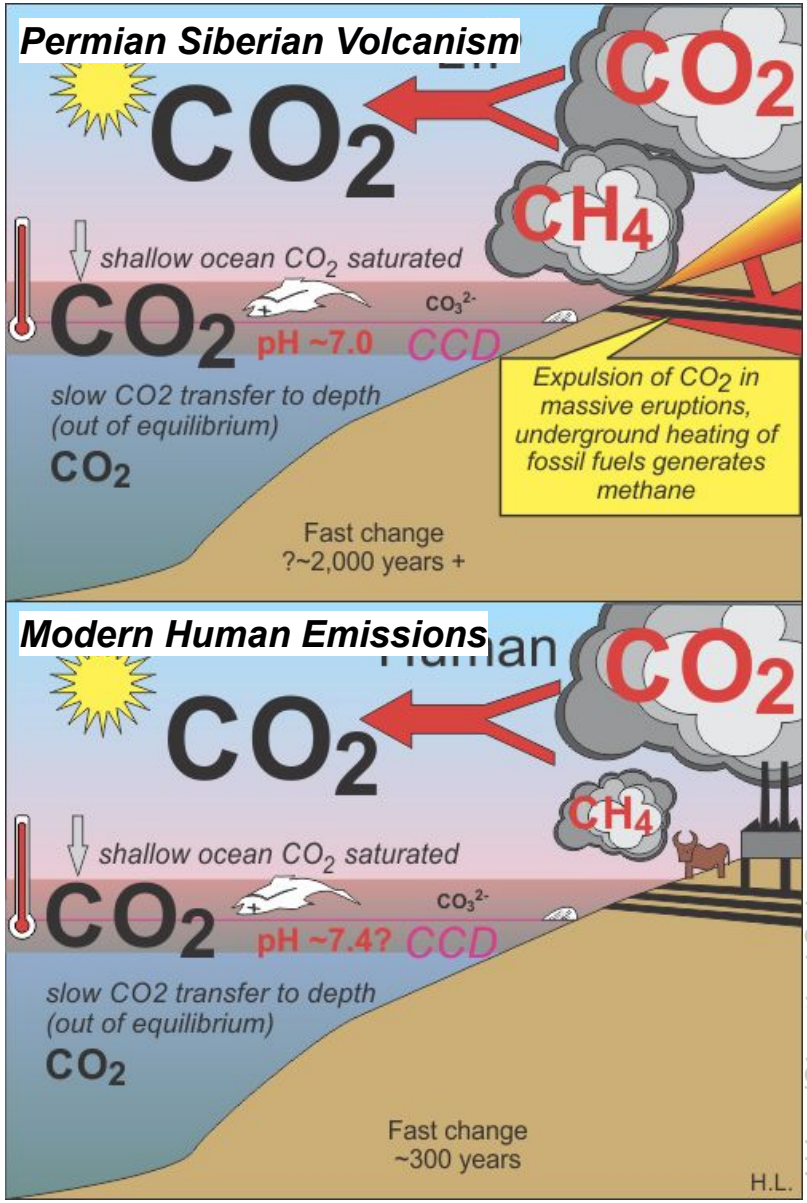
Permian-Triassic Extinction Event

Rugose & Tabulate Corals Calcitic Brachiopods

The biggest victims had **calcareous** skeletons ($CaCO_3$), and sensitive to O_2

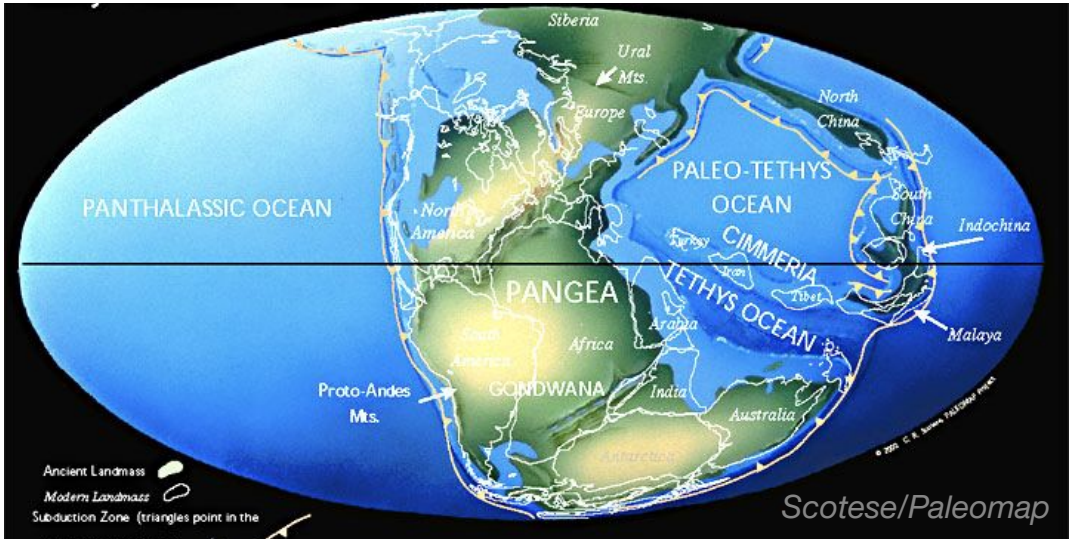


Big spike in global **CO_2** at end Permian
This likely caused ocean acidification, making it difficult for skeletons to form

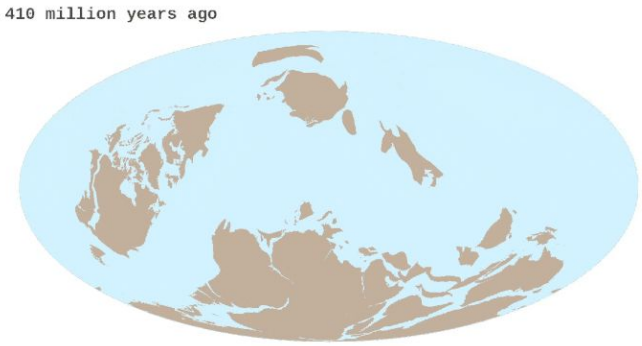


Permian CO_2 related to emissions from Large Igneous Province (LIP)

After the Extinction: Triassic



One giant supercontinent:
Pangaea



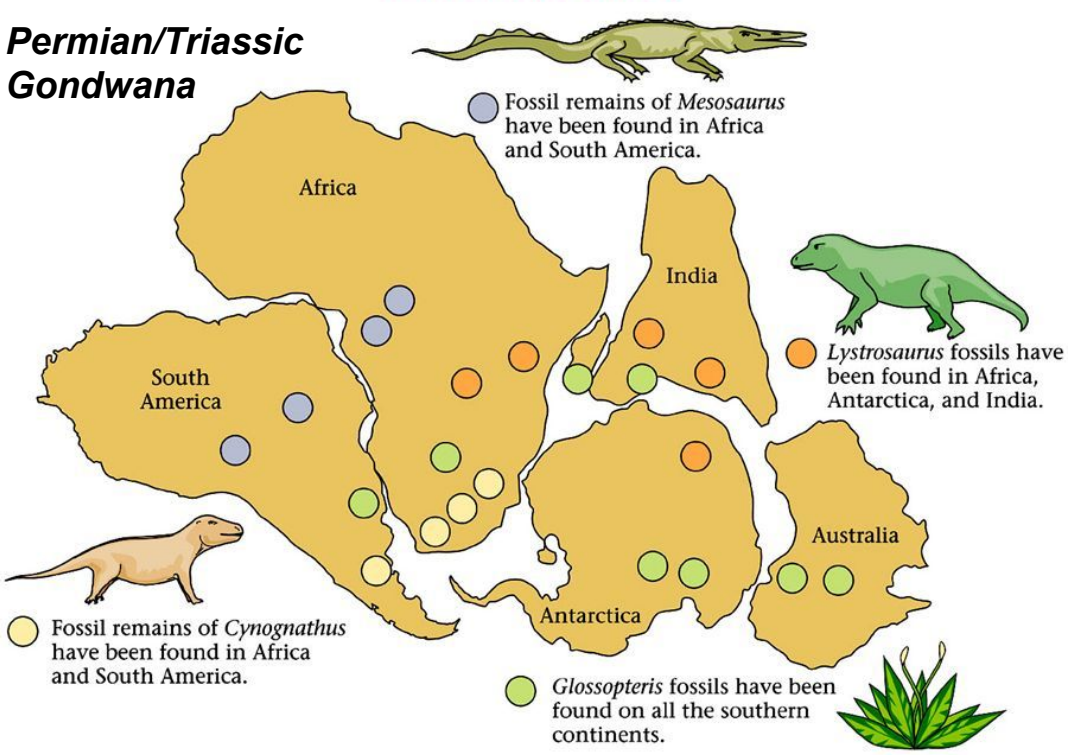
Hot and dry climate, temperate poles

Red bed sandstone

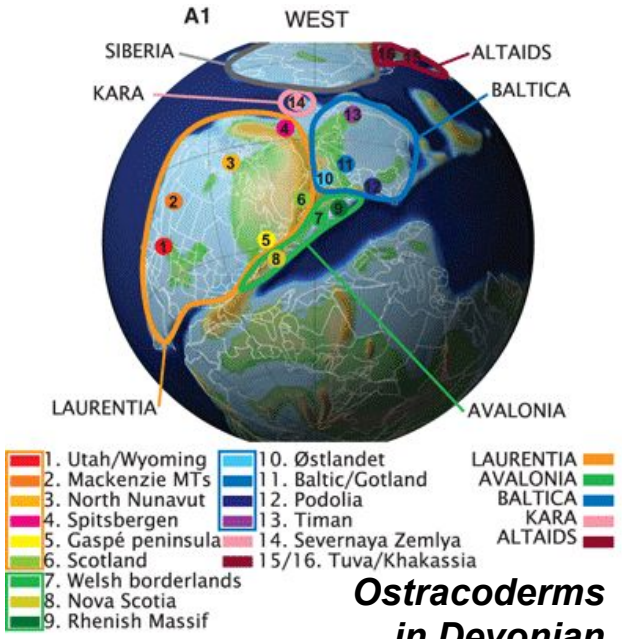
Evaporites

Palaeobiogeography

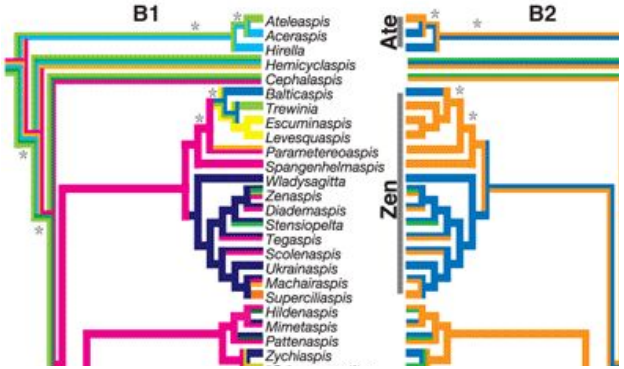
Permian/Triassic Gondwana



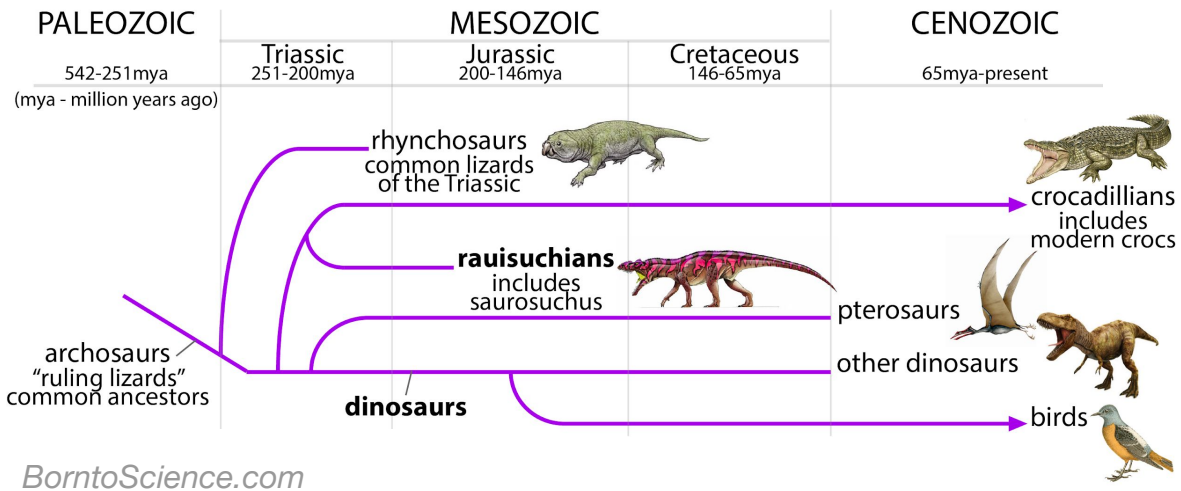
The distribution of fossils was used to support theory of **continental drift** and **plate tectonics** **Palaeobiogeography** informs not only reconstructions of the continents but also the evolution of animals in time and space



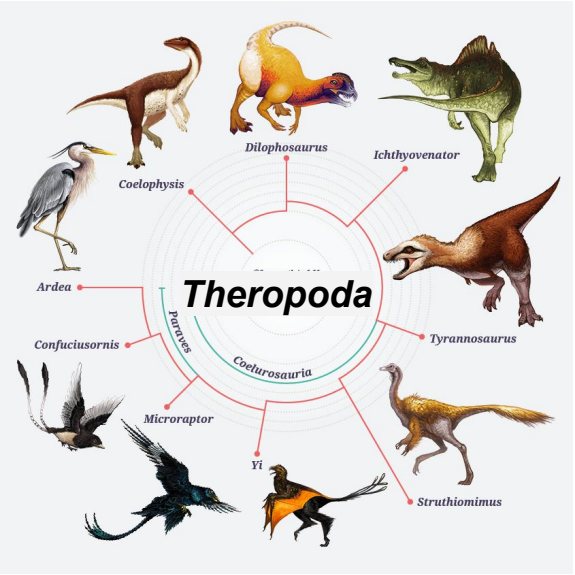
Ostracoderms in Devonian Laurentia

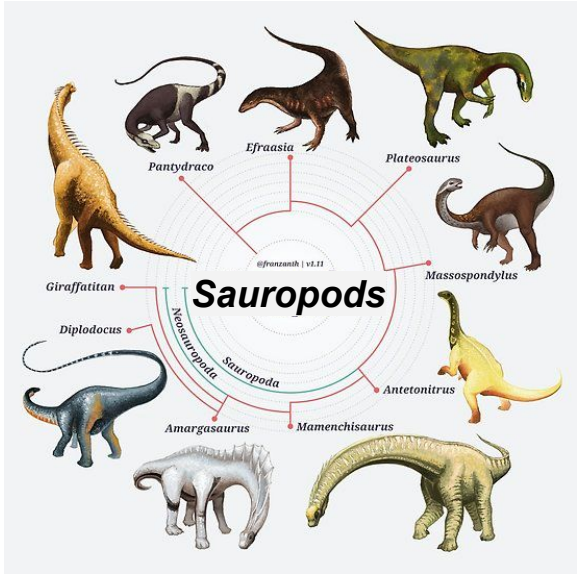


Dinosaurs are Archosaurs

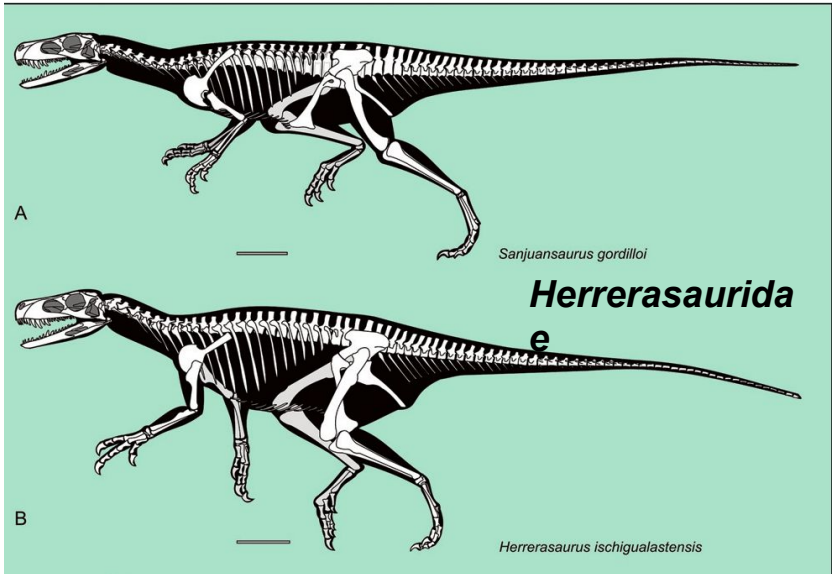


- Birds and crocodiles are **archosaurs**
- Dinosaurs are also archosaurs because they are **stem-birds**
- 3 main clades of dinosaurs





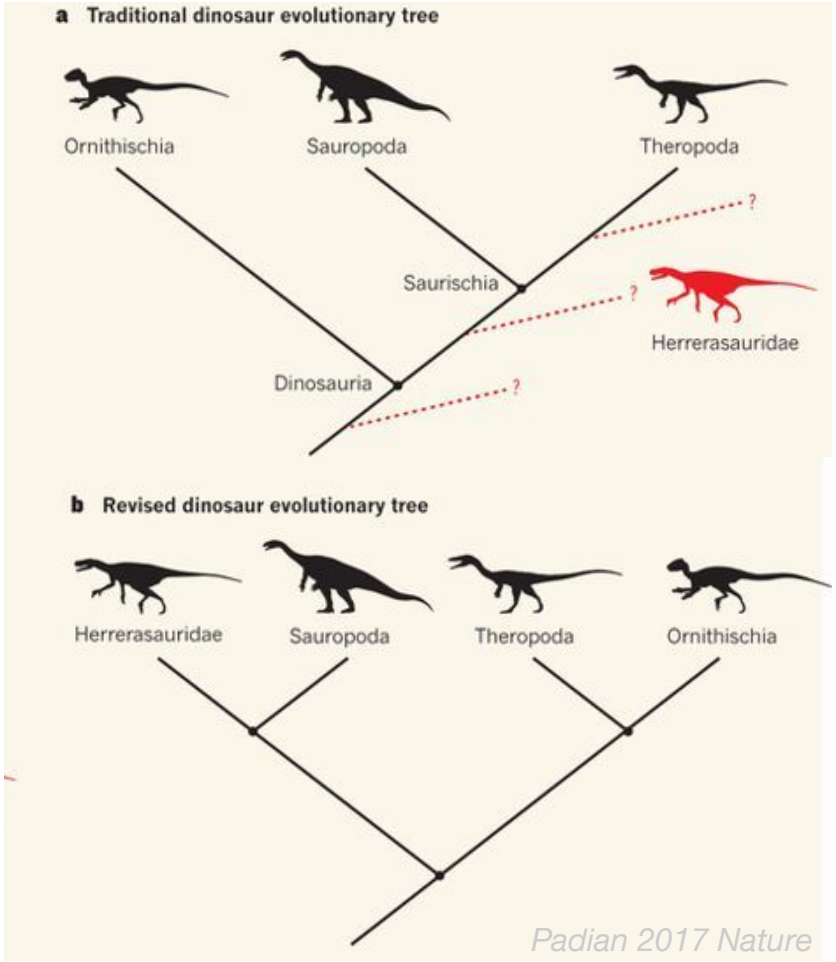
Early Dinosaurs



Alcobar 2010 ZooKeys

Early dinosaurs were small and carnivorous

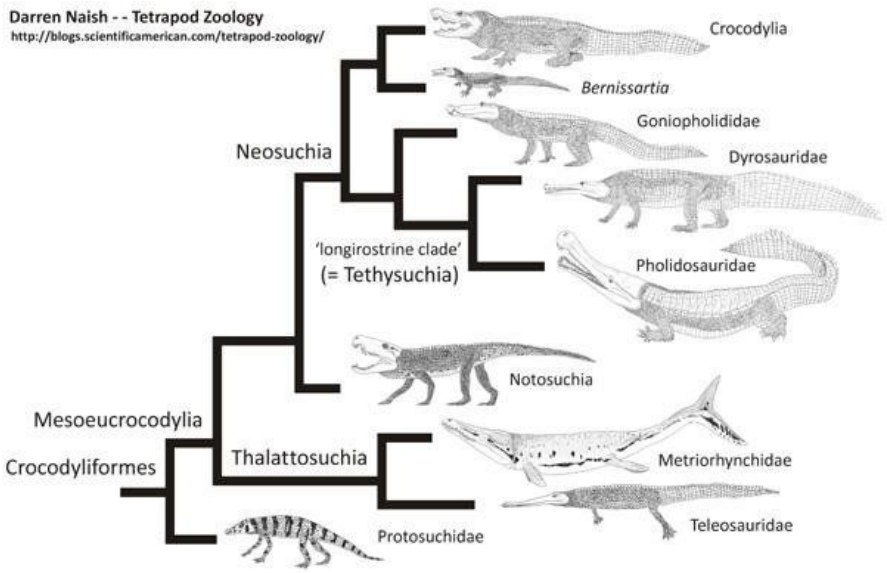
Mixture of advanced and derived traits



Padian 2017 Nature

Recently contentious issue on fundamentals of dinosaur phylogeny

Crocodylomorpha



D Naish

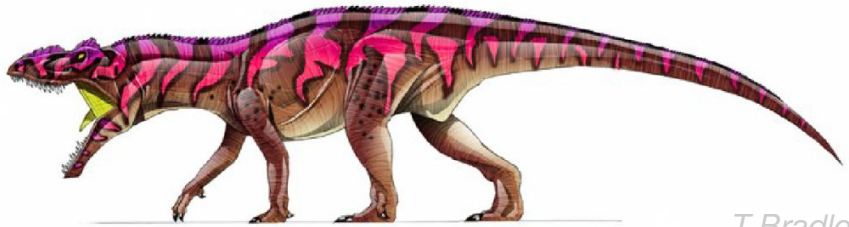
Crocodiles belong to the **Crocodylomorpha**

They had a much bigger diversity in the past than they do today

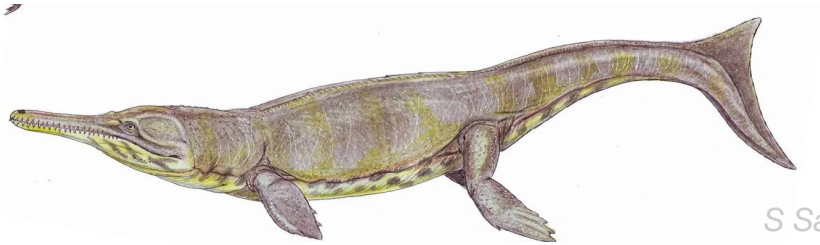
Origins in the Triassic, diversification in the Mesozoic

Modern crocodilians include crocodiles, alligators, gharials and caimans

Mesozoic crocs include giant upright running predators and fully marine swimmers with tail fins



T Bradley

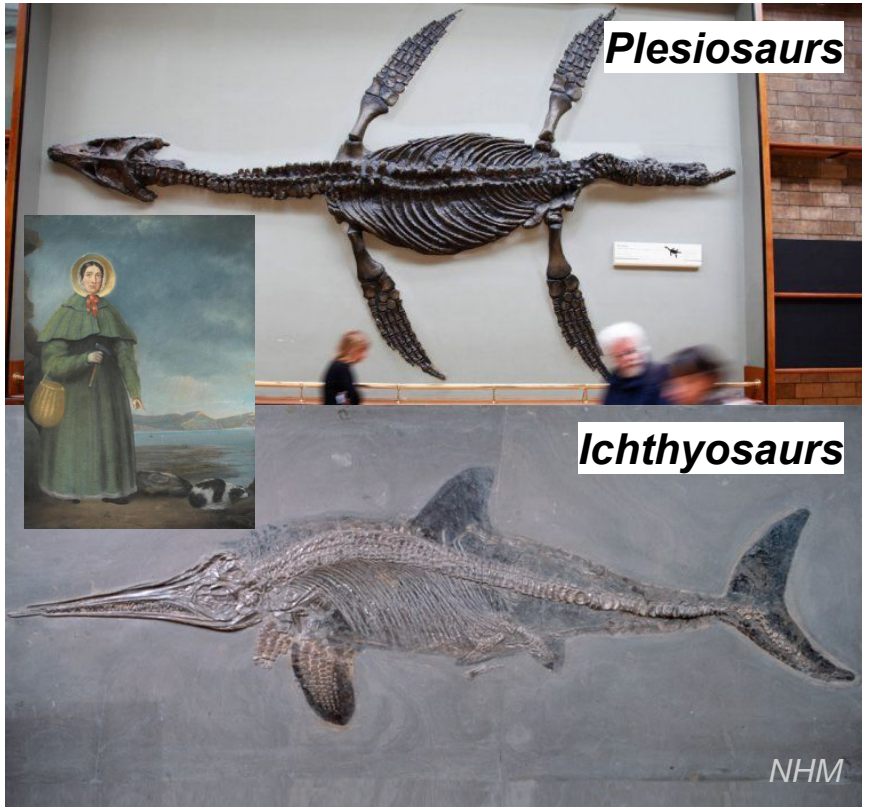


S Sachs

Mesozoic Oceans

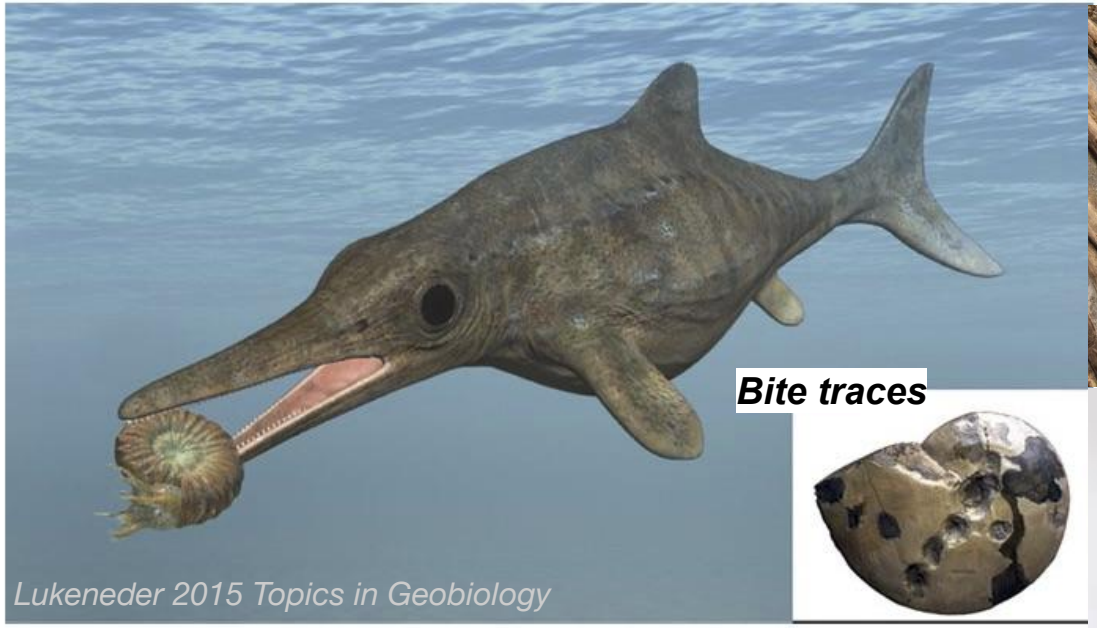
Marine fauna was quicker to bounce back: new corals and fish

Marine Reptiles



Cephalopods

Mesozoic Oceans



Bite traces

Fig. 18.13 Reconstruction of a Lower Jurassic ichthyosaur hunting on ammonoids, biting into the shell of *Dactylioceras* in the epipelagic water zone of the Liassic sea (based on studies of Posidonia shales in SW Germany). Upper Cretaceous mosasaur bite marks (max. puncture diameter 3 cm; lower right corner) on a Campanian *Placenticeras* shell (cast replica, max. diameter 30 cm) from the Western Interior Seaway in South Dakota

Belemite Gut contents of an ichthyosaur

Marine reptile coprolite

Evidence of predation in the fossil record can come from bite marks, gut contents and coprolites (fossil poo)

Dinosaurs!
7.1 Triassic

Intended Learning Objective:
Compare the Permian-Triassic and Cretaceous-Palaeogene mass extinction events in terms of their possible causes and their impacts on life on earth

Now take a look at the associated task/question for this section

Robert Sansom, U. Manchester

Part 7
Dinosaurs!

7.2 Jurassic

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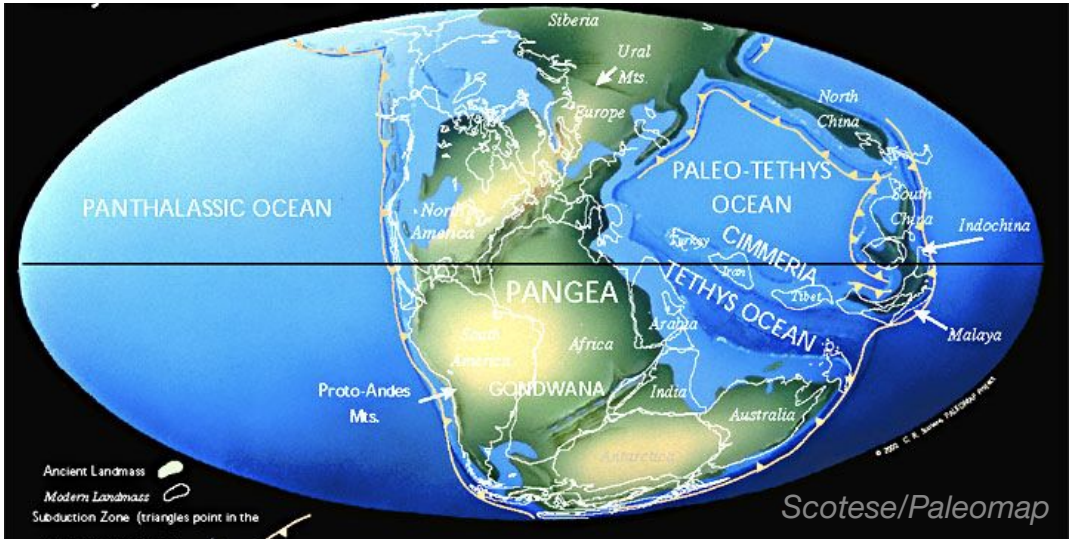
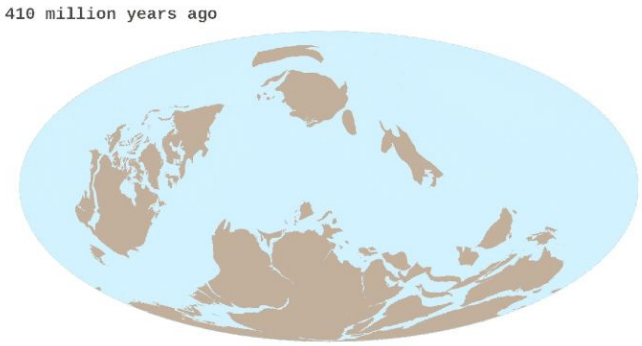
Robert Sansom
University of Manchester

Triassic



One giant supercontinent:
Pangaea

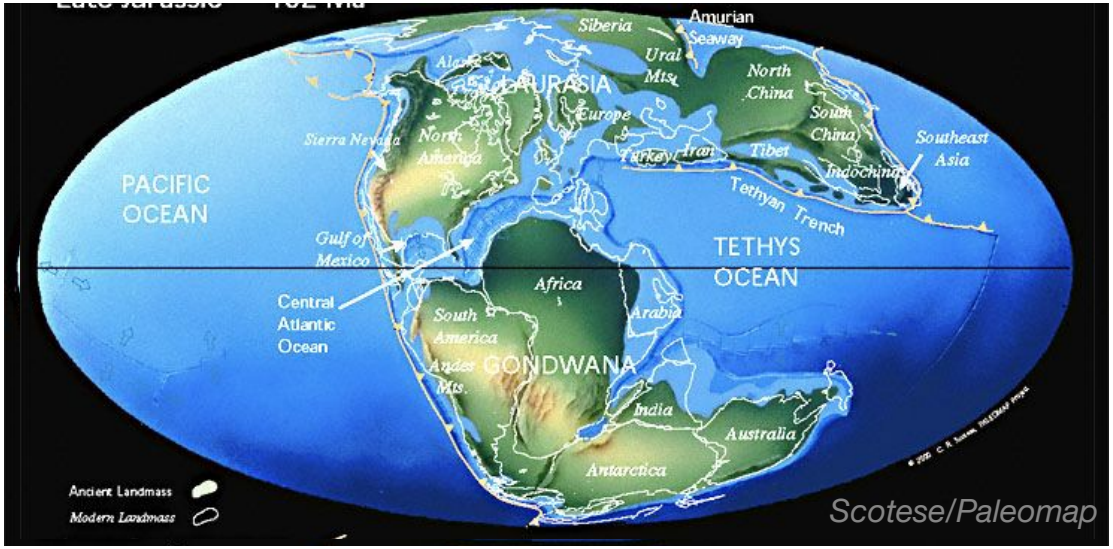


Hot and dry climate, temperate poles

Red bed sandstone

Evaporites

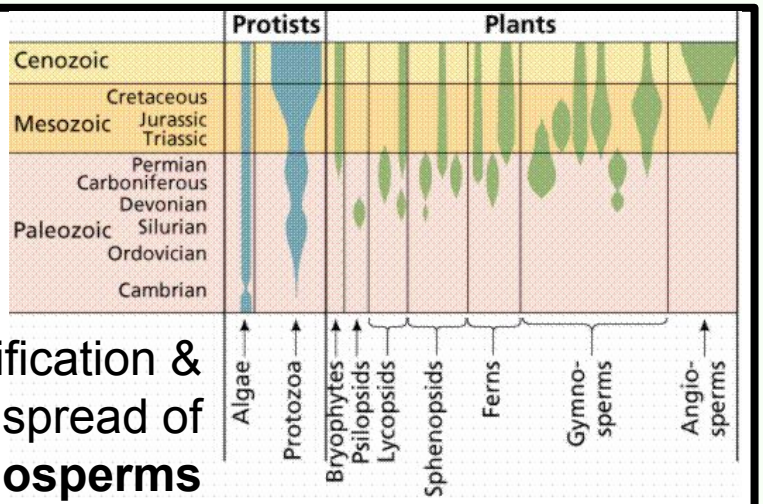
Jurassic



Pangaea supercontinent beginnings to split into two: **Laurasia** in the north and **Gondwana** in the south

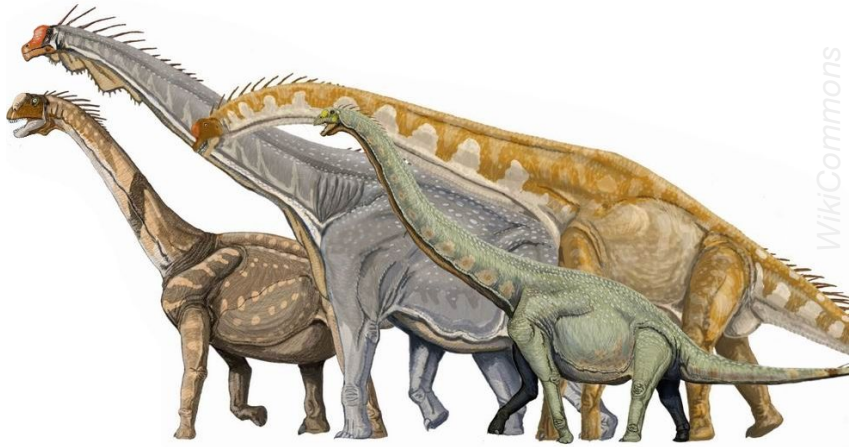
Also, closure of the Tethys ocean begins

More humid and subtropical conditions



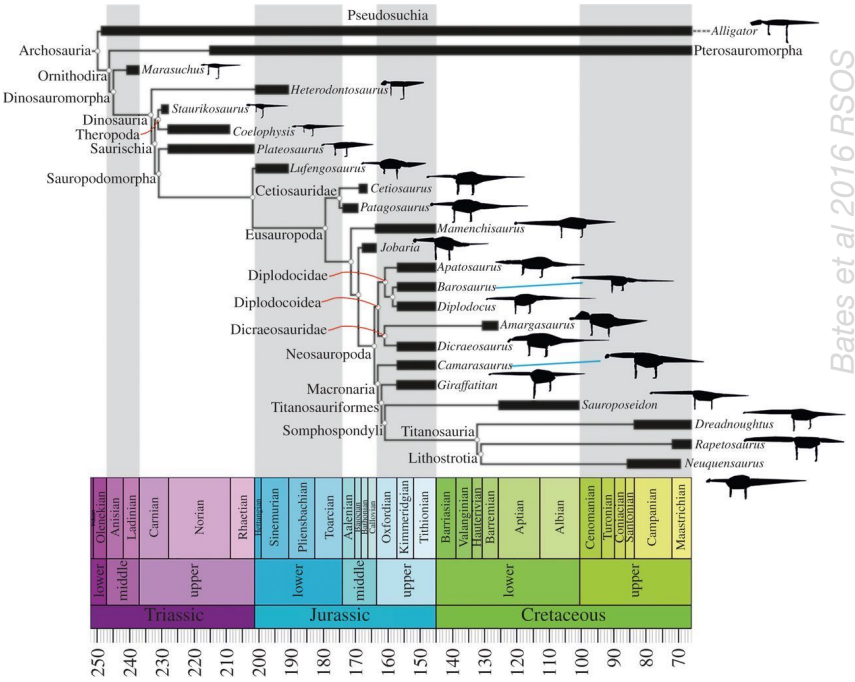
Diversification & spread of **gymnosperms** e.g. conifers

Sauropods

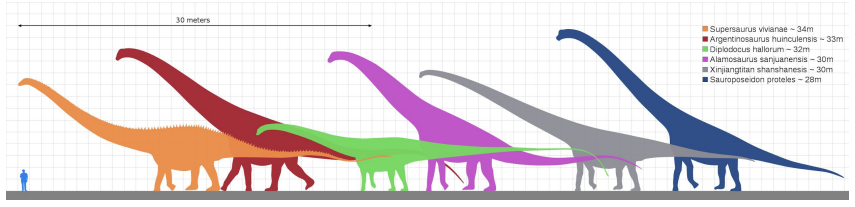


Sauropods have long necks, long tails & small heads

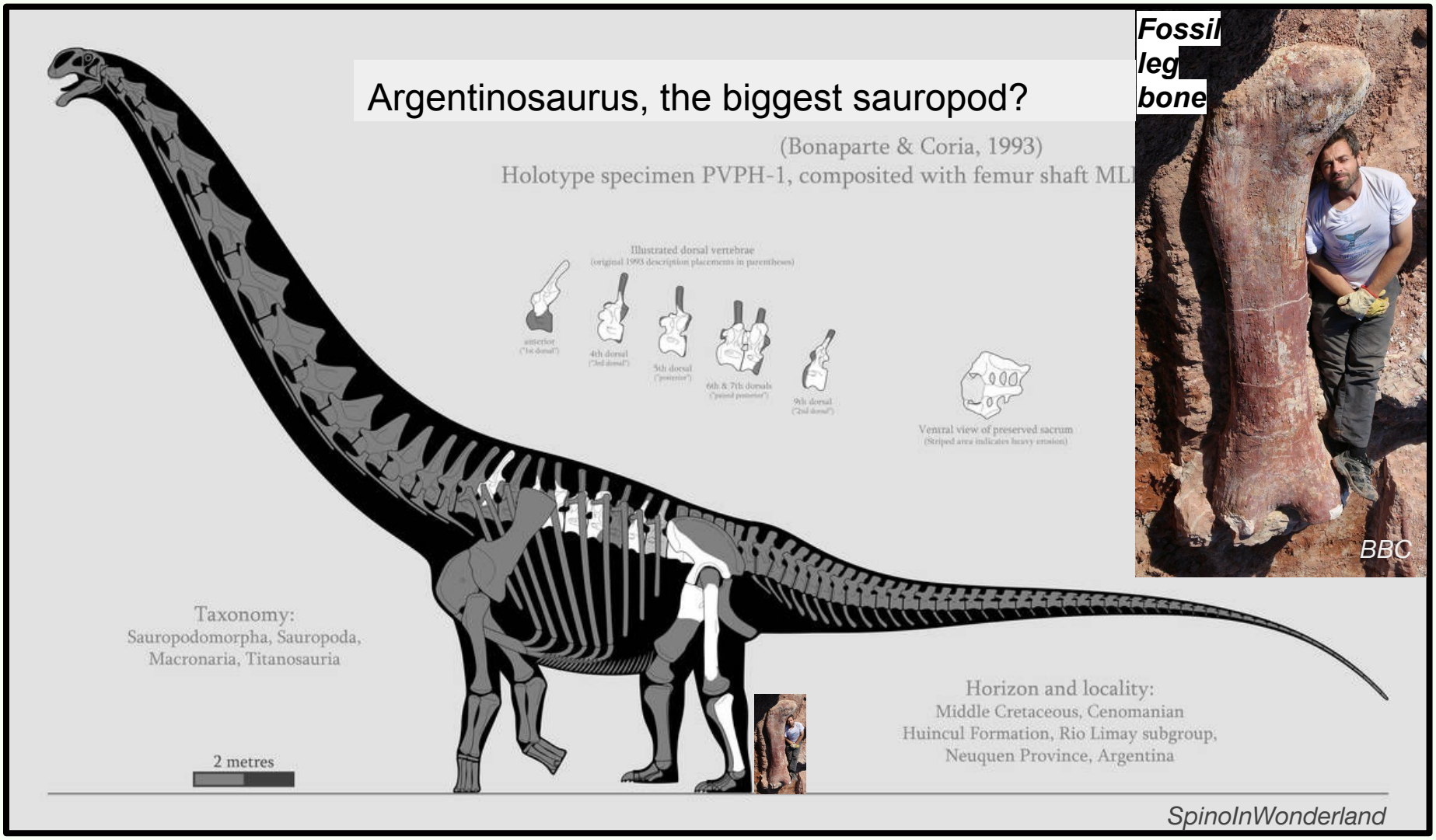
Adaptations to reach high foliage



They originated in the Triassic, diversified in Jurassic, progressively increasing **body size**



Sauropods


Argentinosaurus, the biggest sauropod?
(Bonaparte & Coria, 1993)
Holotype specimen PVPH-1, composited with femur shaft ML
Illustrated dorsal vertebrae
(original 1993 description placements in parentheses)
anterior
("1st dorsal")
4th dorsal
("3rd dorsal")
5th dorsal
("posterior")
6th & 7th dorsals
("paired posterior")
9th dorsal
("2nd dorsal")
Ventral view of preserved sacrum
(Striped area indicates heavy erosion)
Taxonomy:
Sauropodomorpha, Sauropoda,
Macronaria, Titanosauria
2 metres
Horizon and locality:
Middle Cretaceous, Cenomanian
Huincul Formation, Rio Limay subgroup,
Neuquen Province, Argentina
SpinoInWonderland

Fossil leg bone
BBC

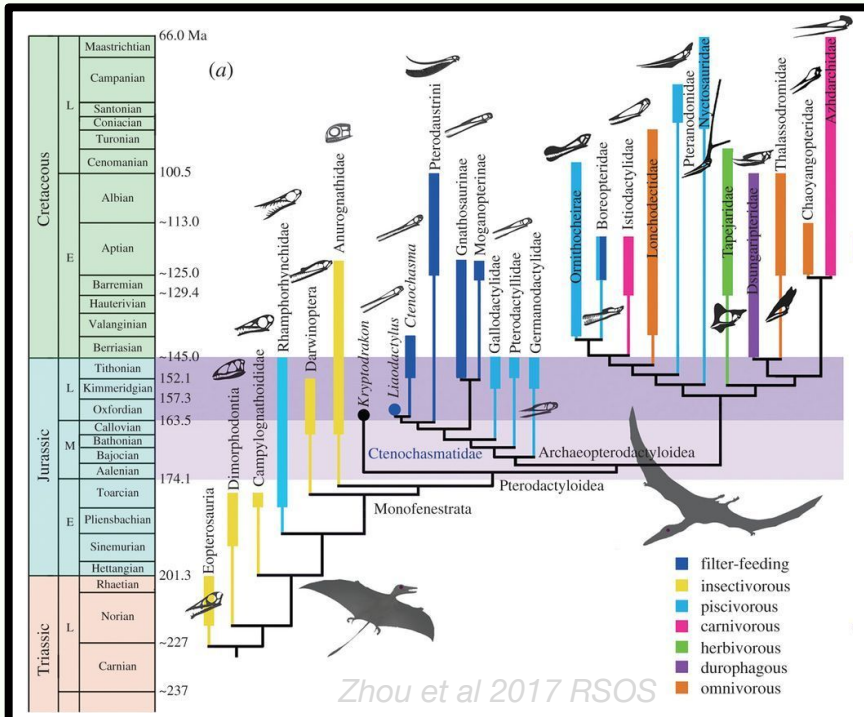
Pterosaurs

AMNH

Yang et al 1019 Nature EE

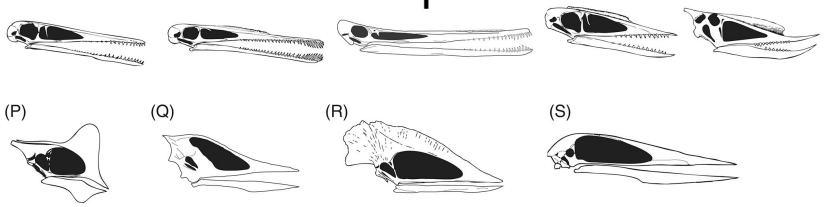
Pterosaurs are flying reptiles (“winged lizards”), **not** dinosaurs

Hollow bones & fluffy ‘proto-feathers’



Zhou et al 2017 RSOS

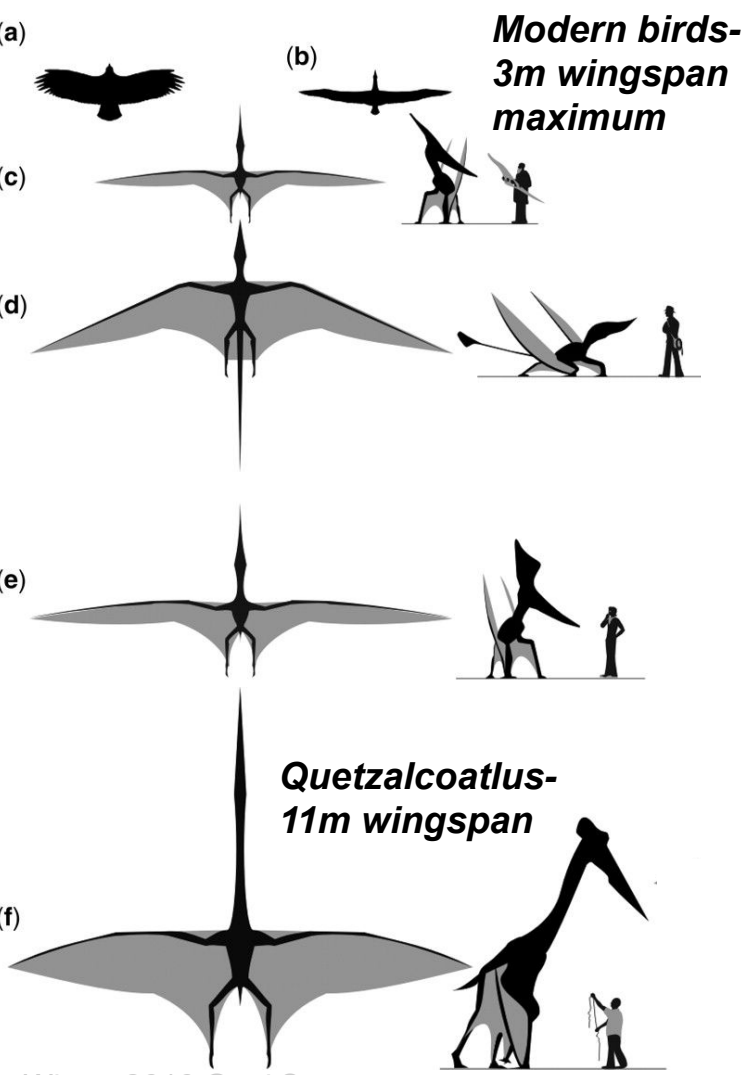
Pterosaurs also originated in the Triassic and diversified in Jurassic
Wide range of diets, mostly insectivorous and piscivorous



Bestwick et al Biol Rev

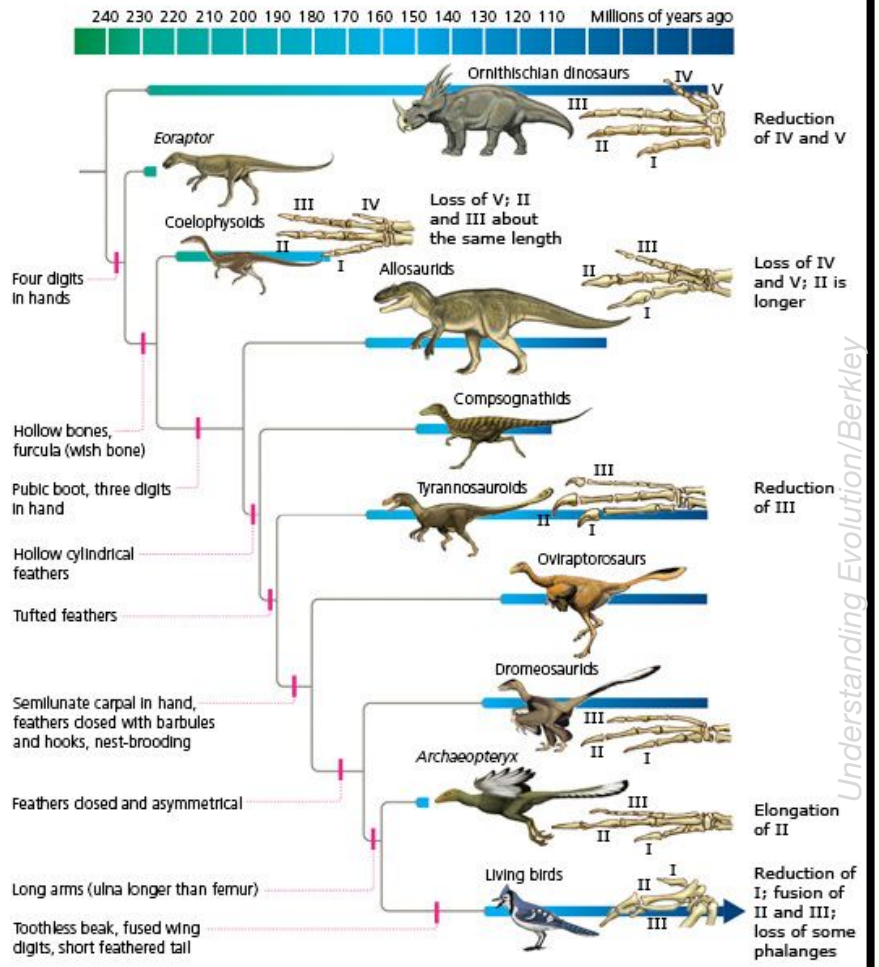
Pterosaurs

Quetzalcoatlus


(a)
(b)
Modern birds-
3m wingspan
maximum
(c)
(d)
(e)
Quetzalcoatlus-
11m wingspan
(f)
Witton 2010 Geol Soc

Theropods

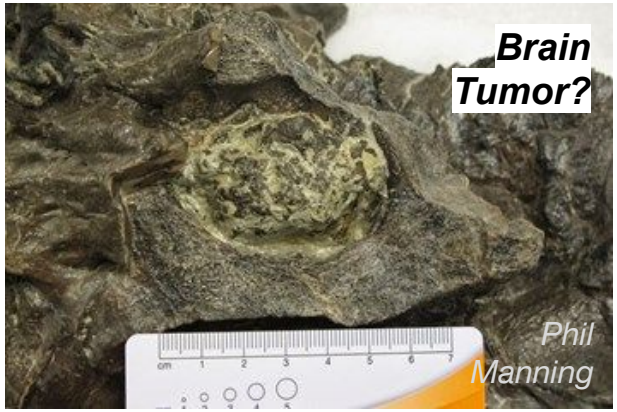
Theropods are a diverse dinosaur group that includes **birds**
They have three-toed limbs and feathers, were mostly carnivorous



Theropods also originated in the Triassic, and diversified later

Theropods

Our own ***Gorgosaurus***, a classic theropod closely related to *Tyrannosaurus rex*, exhibits interesting pathologies



Broken and healed femur

Indianapolis Children's Museum

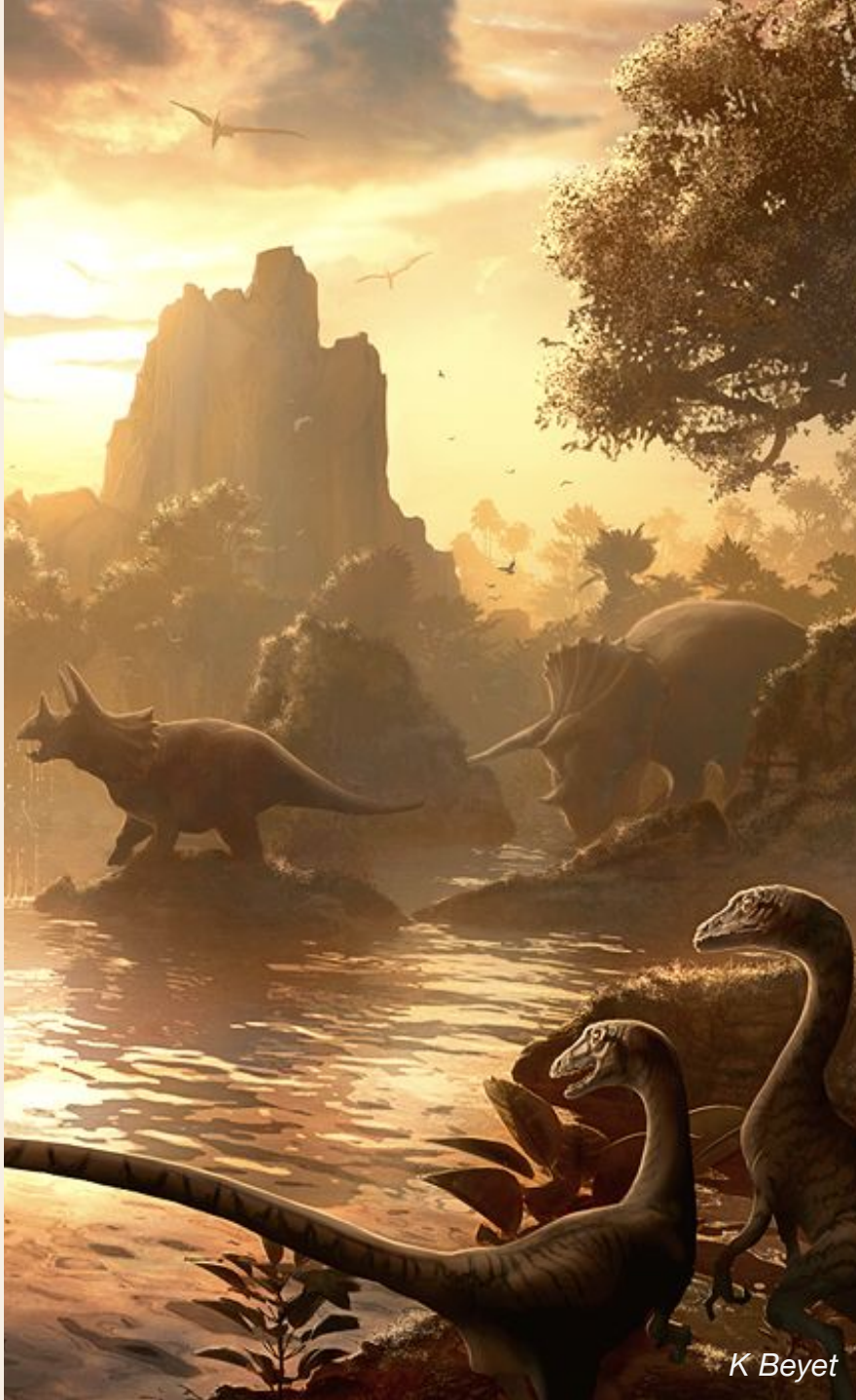
https://twitter.com/evopalaeo/status/1009509371217436672?s=20

Dinosaurs!
7.2 Jurassic

Intended Learning Objective: Define the major clades of extinct and extant archosaurs, describe their diversity through time, and relate that to geological events during the Mesozoic

Now take a look at the associated task/question for this section

Robert Sansom, U. Manchester

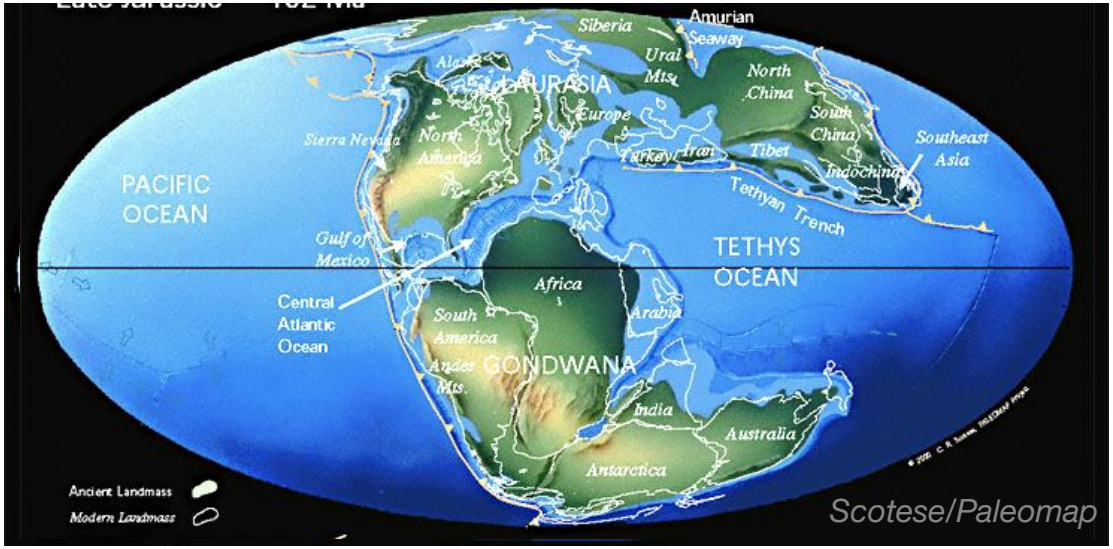


Part 7
Dinosaurs!

7.3 Cretaceous

EART11100: How to Grow a Planet
Robert Sansom
University of Manchester

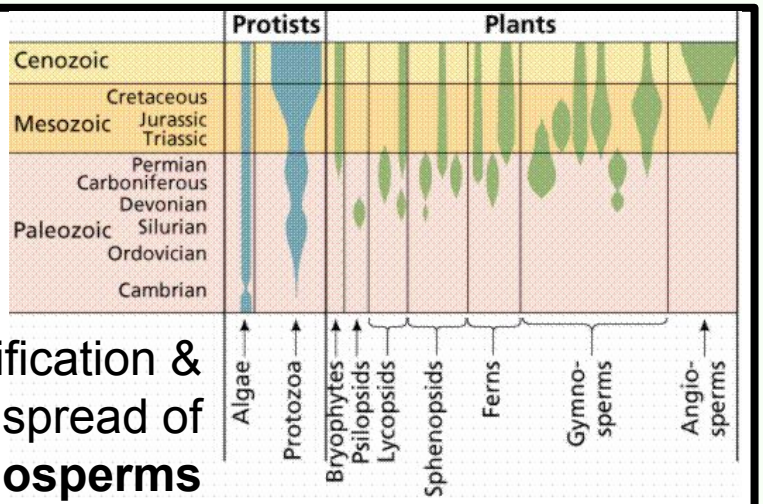
Jurassic



Pangaea supercontinent beginnings to split into two: **Laurasia** in the north and **Gondwana** in the south

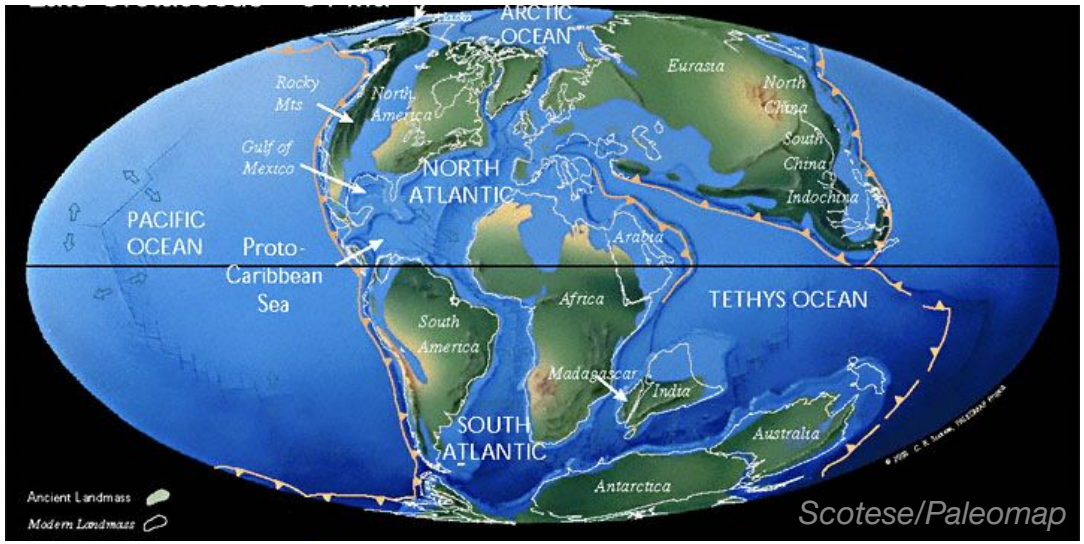
Also, closure of the Tethys ocean begins

More humid and subtropical conditions

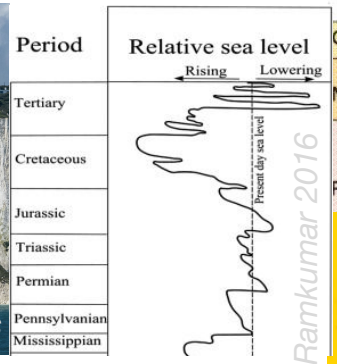


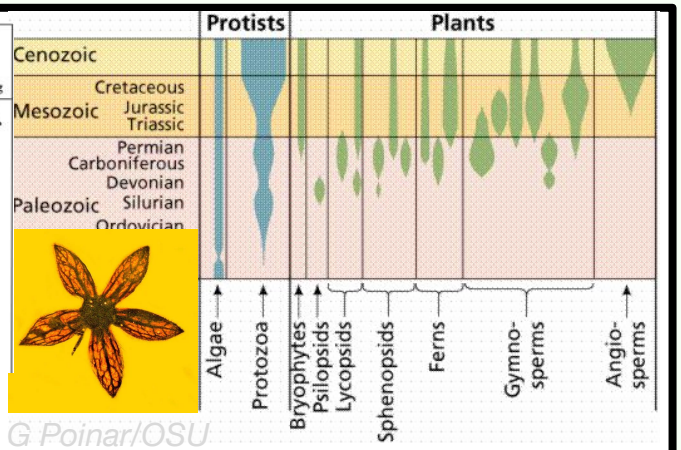
Diversification & spread of **gymnosperms** e.g. conifers

Cretaceous



Laurasia splits into North America and Eurasia
Gondwana splits into South America, Africa, Antarctica, Australasia, and India
Opening of Atlantic Ocean, closure of Tethys

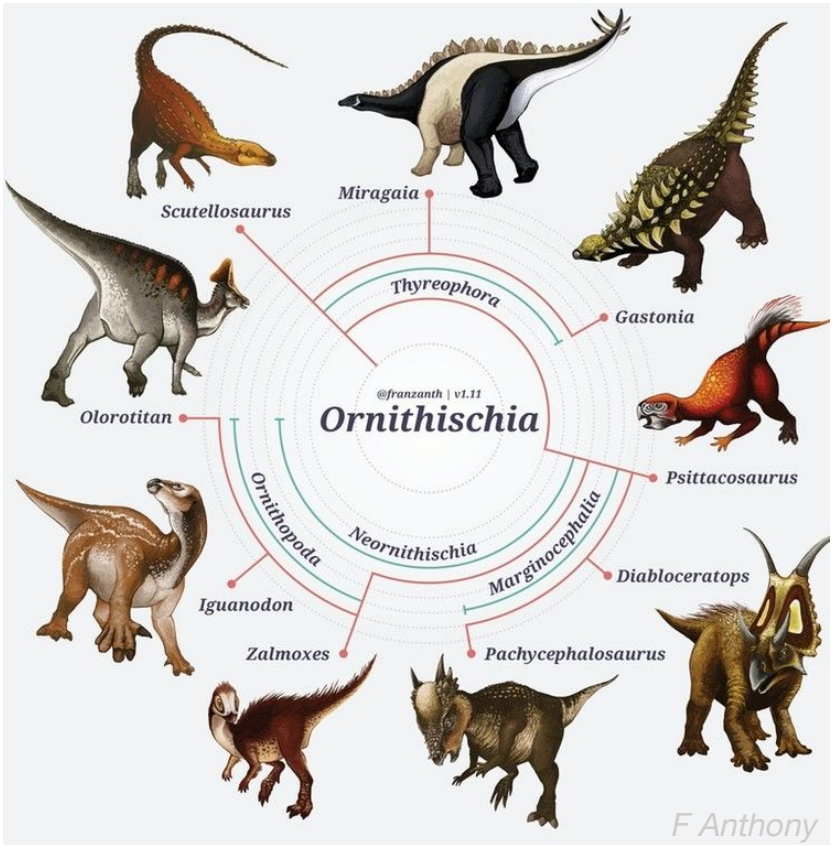




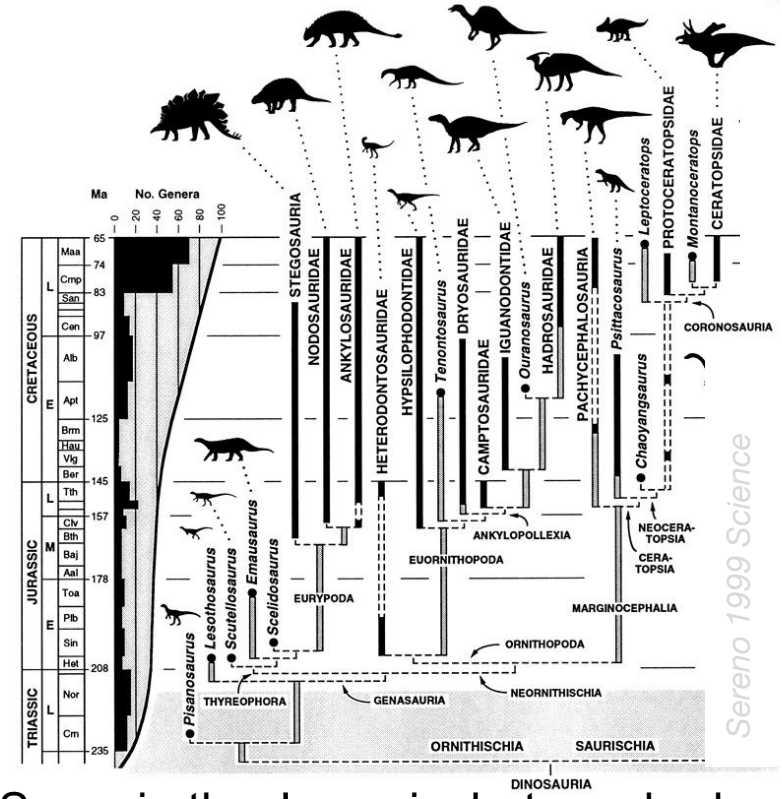
Warm climate, high sea levels
Extensive inland seas meant high sedimentary deposition, particular limestones (e.g. chalk)

Diversification and spread of **angiosperms** e.g. flowering

Ornithischians

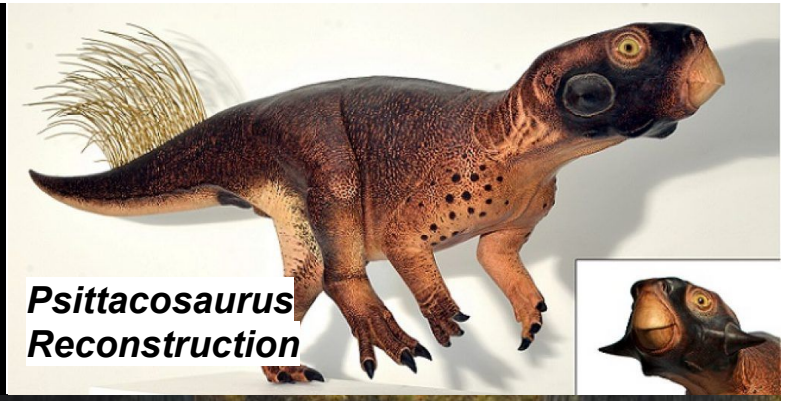


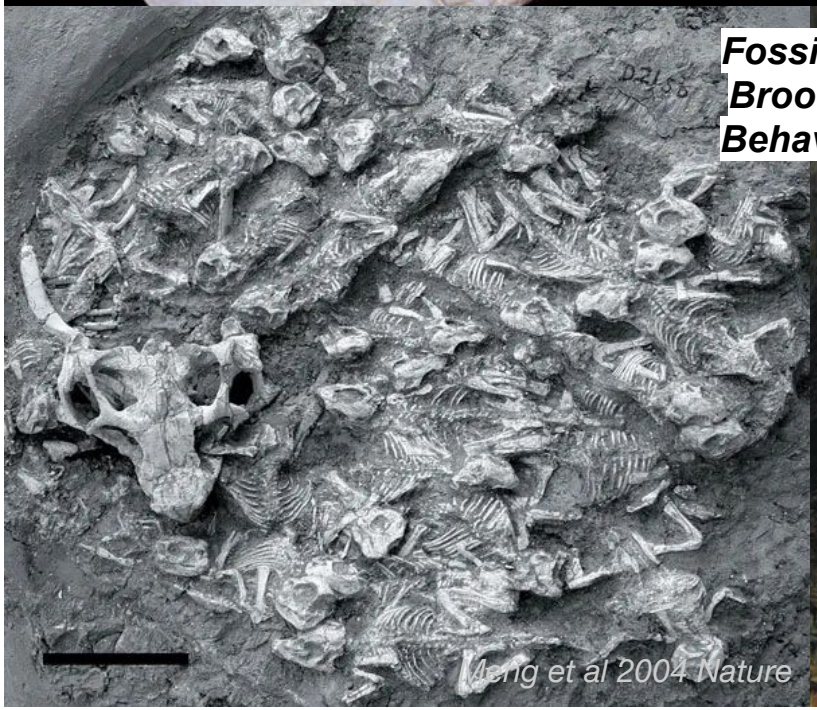
Ornithischia (‘bird-hipped’), a diverse group of dinosaurs with ‘beak’ like jaw
Most ornithischians are herbivorous, and mostly quadrupedal



Some in the Jurassic, but reached peak diversity in the Cretaceous

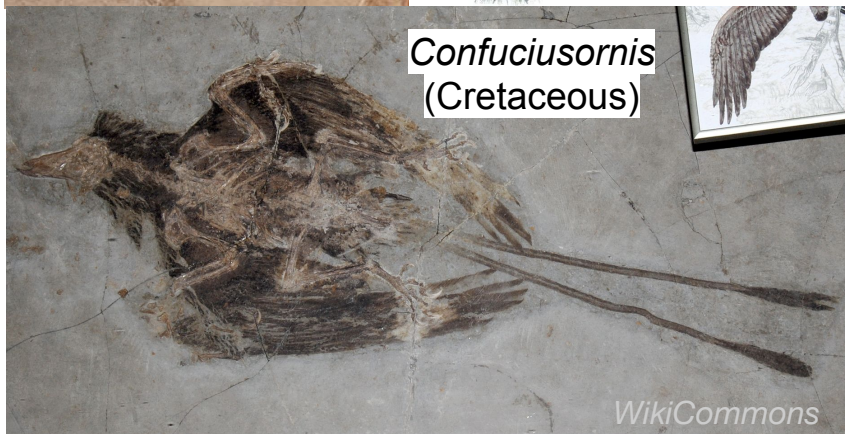
Ornithischians





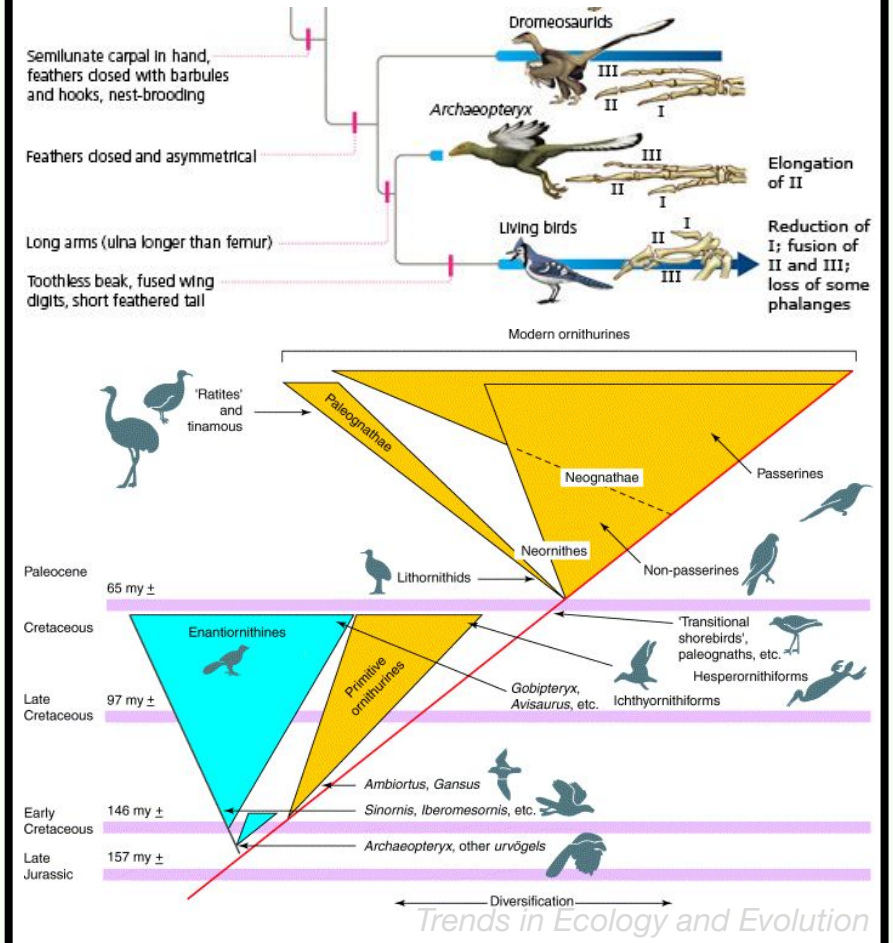
Avialae (flying theropods)

Archaeopteryx
(Jurassic)



Confuciusornis
(Cretaceous)

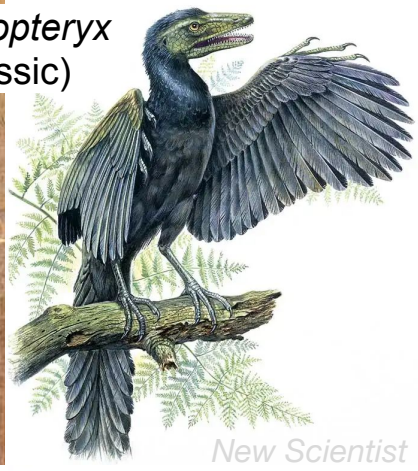
Avialae are theropods that fly, including **birds**
True feathers and powered wings,



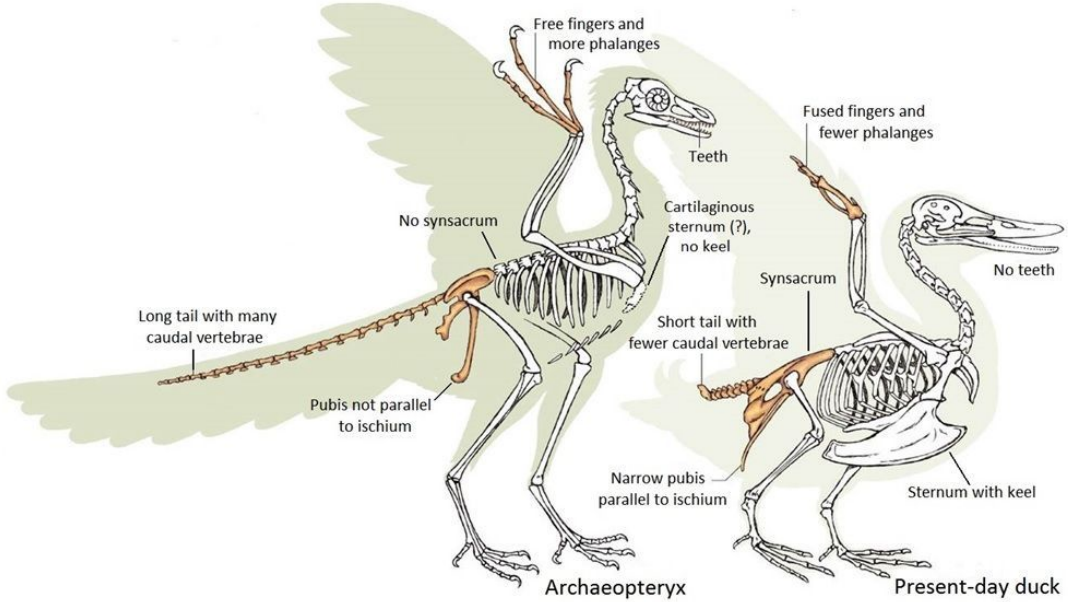
Avialae evolved within theropods, first diversified during the Cretaceous, but most birds evolved after that

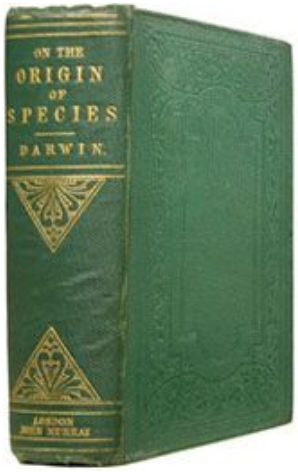
Avialae (flying theropods)

Archaeopteryx
(Jurassic)



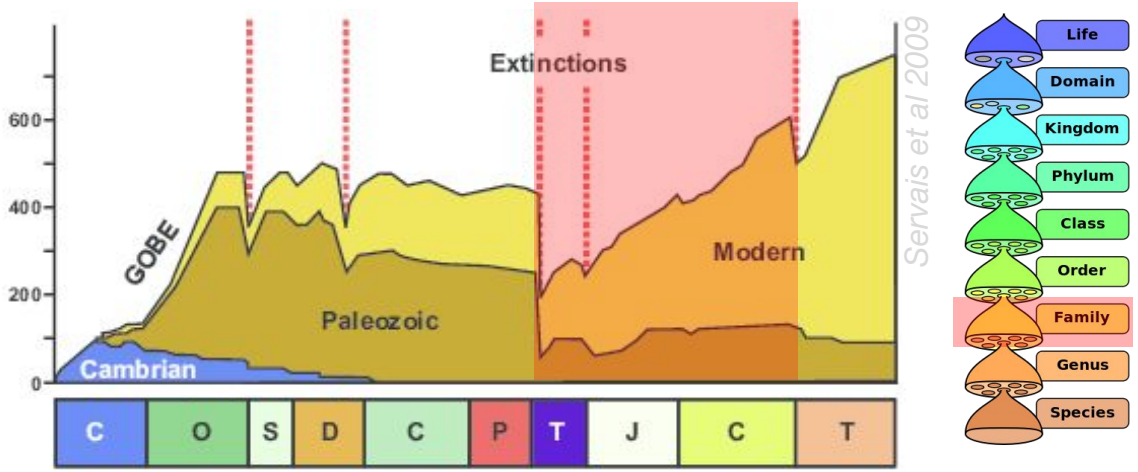
Archaeopteryx is a great example of a **transitional fossil**.
It exhibits avian features: feathers, wings, and theropod features: teeth, long tail with bones, claws.
It was described 2 years after Darwin's origin of species, lending strong support to evolutionary theory

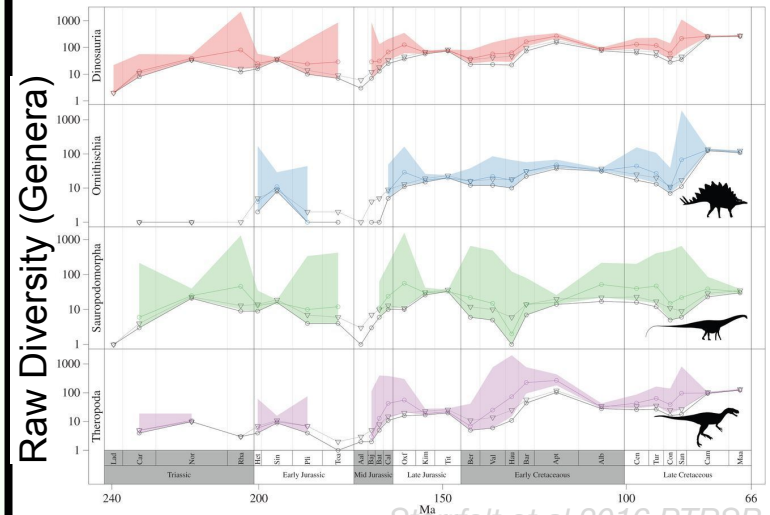




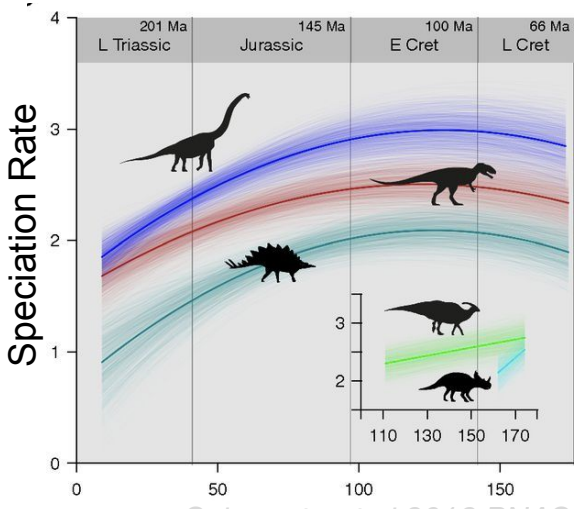
Mesozoic Diversity Patterns

After the Permian-Triassic extinction, ‘Palaeozoic’ marine faunas remained minor, whilst ‘modern’ marine faunas diversified and became more dominant





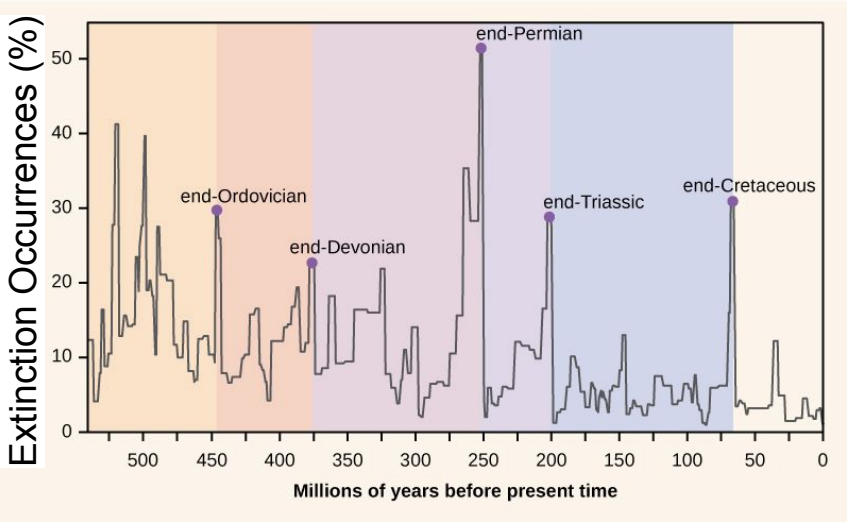
Starrfelt et al 2016 PTRSB



Sakamoto et al 2016 PNAS

Raw dinosaur data also show increasing diversity during throughout the Mesozoic
But statistical modelling indicates they were in decline by the Cretaceous

Cretaceous-Palaeogene Extinction Event

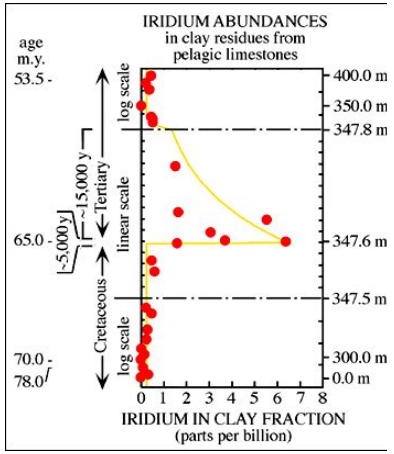


Openstax Biology

WikiCommons

There was a big faunal change from Cretaceous to Cenozoic
End of dinosaurs (but not birds), ichthyosaurs, pterosaurs, ammonoids, many others affected

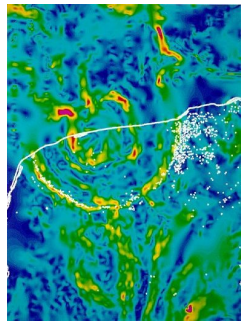
Rhodri Jerrett, U of Manchester



Alvarez et al 1980

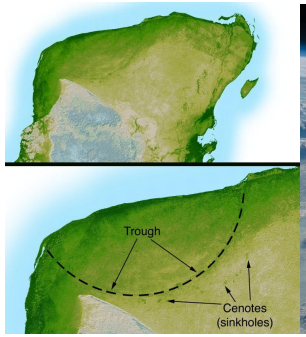
The Cretaceous-Palaeogene boundary shows big geological changes, including a huge **Iridium spike**

Pearson Education



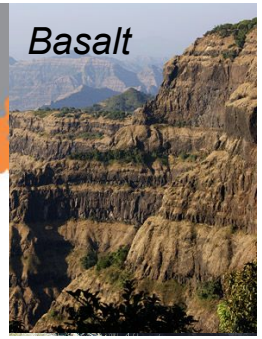
Matched to massive **Chicxulub crater** impact in Yucutan, Mexico

Cretaceous-Palaeogene Extinction Event

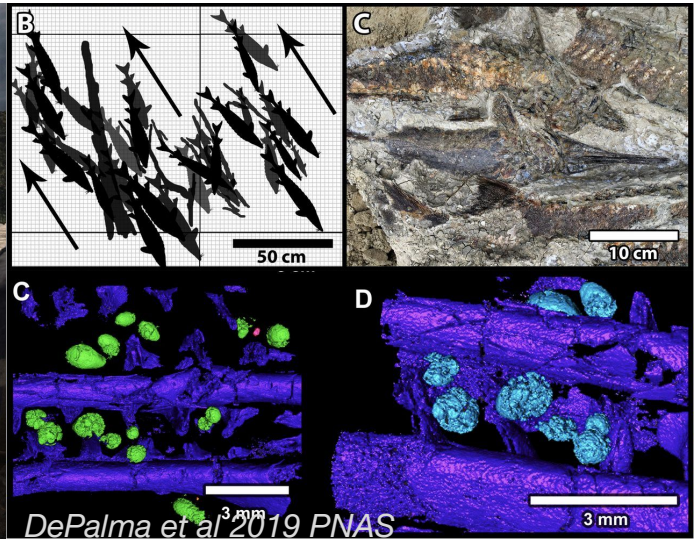


Massive extra-terrestrial asteroid impact: **Chicxulub crater,** Mexico

OR

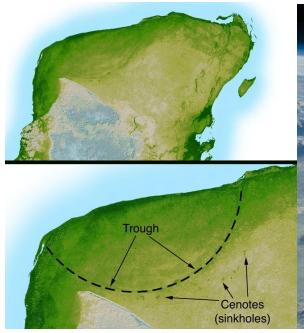


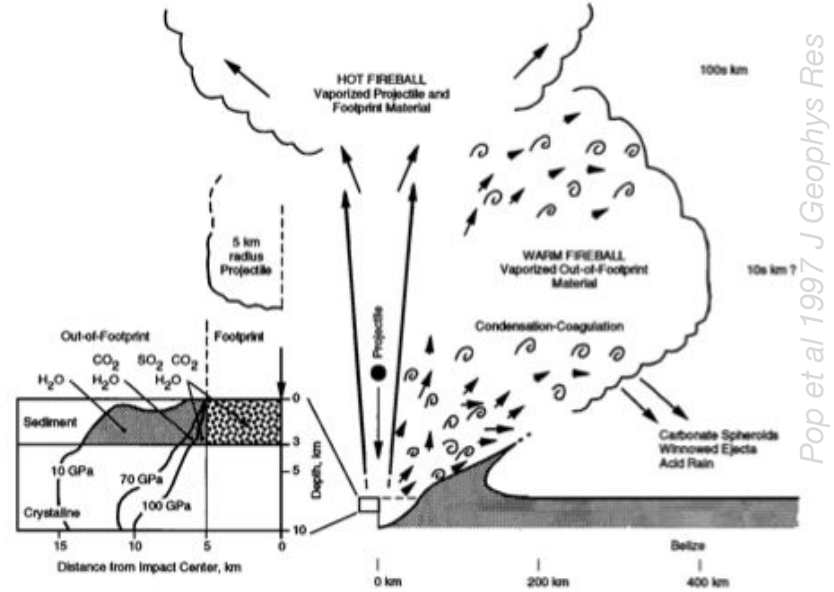
Massive volcanic eruptions, flows, and gases: **Deccan Traps,** India



New site in Dakota finds fish with ejecta, preserved in ‘snapshot’ following the impact

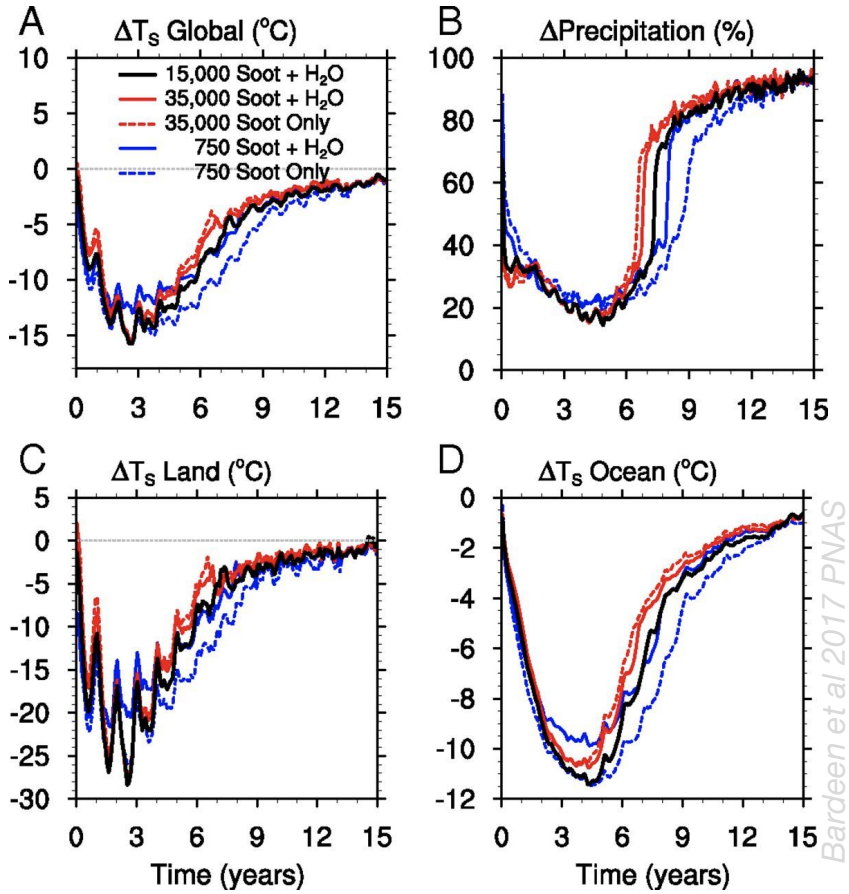
Cretaceous-Palaeogene Extinction Event





Pop et al 1997 J Geophys Res

The impact ejected large amounts of dust, soot, and sulphate aerosols into the atmosphere



Bardeen et al 2017 PNAS

Modelling indicates that it would have caused a drastic **climatic cooling** (‘nuclear winter’) for years after

Mesozoic summary

- Life recovers from the End Permian mass extinction event during the relatively dry Triassic
- Massive diversification of animals and plants on land in the Jurassic
- Dinosaurs ‘slow down’ during the Cretaceous before meeting their fate in the End-Cretaceous impact mass extinction event

Dinosaurs!
7.3 Cretaceous

Intended Learning Objective: Compare the Permian-Triassic and Cretaceous-Palaeogene mass extinction events in terms of their possible causes and their impacts on life on earth

Now take a look at the associated task/question for this section

Robert Sansom, U. Manchester

Part 7
Dinosaurs!

7.4 The Real Jurassic Park: Engineering

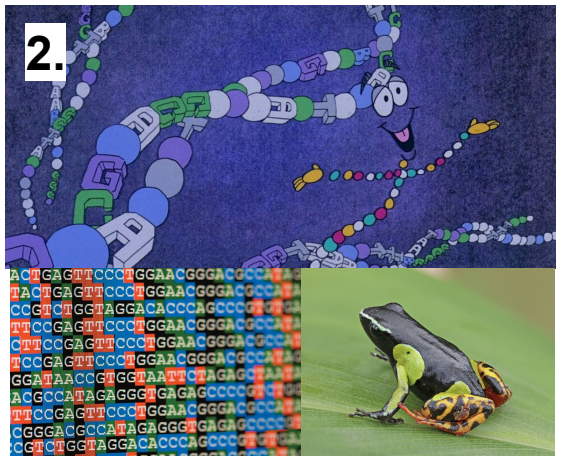
EART11100: How to Grow a Planet
Robert Sansom
University of Manchester

The Premise of "*Jurassic Park*"

1.

Dinosaur blood from mosquitoes in amber

2.



Dinosaur DNA extracted, combined with frog DNA

3.

DNA sequence used to rear dinosaurs

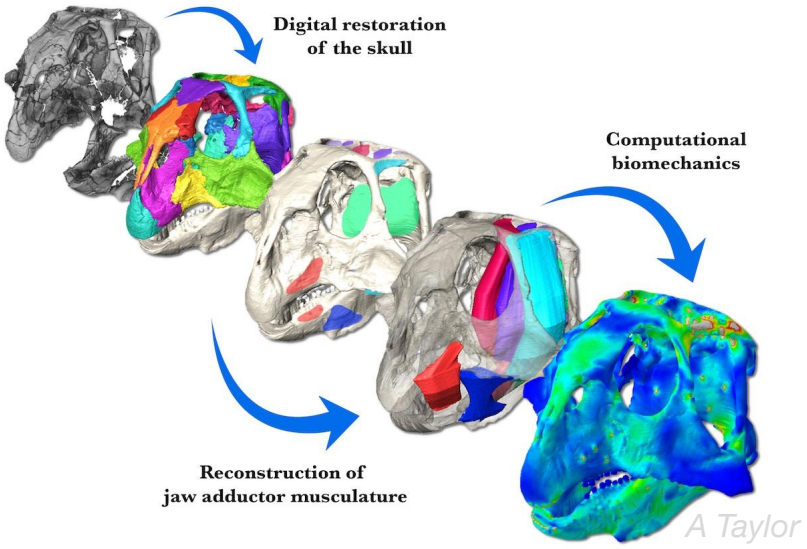
But… DNA is far too fragile and does not survive anywhere near that long (66 million years+)

No DNA from copal (10,000 years ago)

Oldest recovered DNA is 800,000 years old

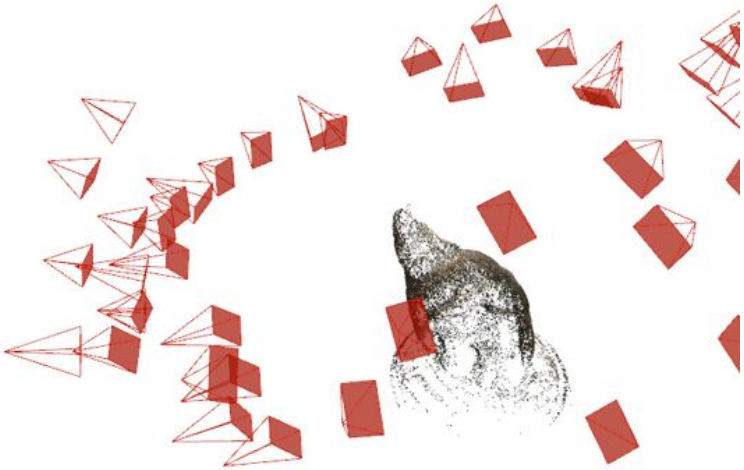
How can we bring the dead back to life?

Engineering approaches can help us reconstruct the motion and behaviour of long extinct animals, for example feeding and locomotion



Studying exceptionally preserved **soft tissues** can provide insights into appearance and behaviour of extinct animals, for example colours

External Scanning: Photogrammetry and LiDAR



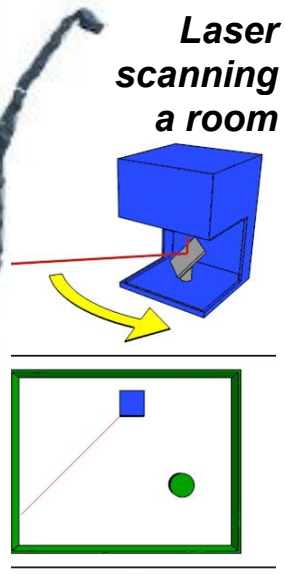
P Falkingham

ESI/U. Witwatersrand/Morphosource

Photogrammetry uses 2D images from multiple angles to create a virtual 3D model

University of Manchester

Digitally scanning skeletons and reconstructing bodies

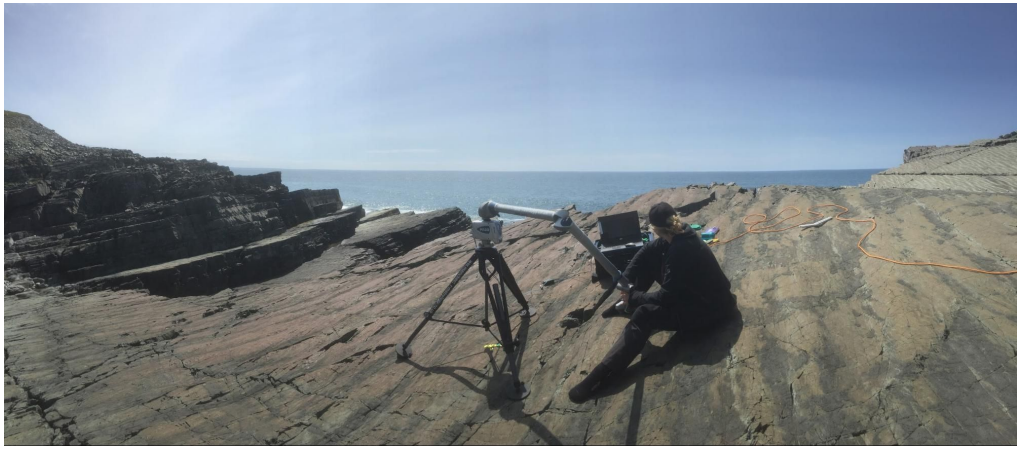


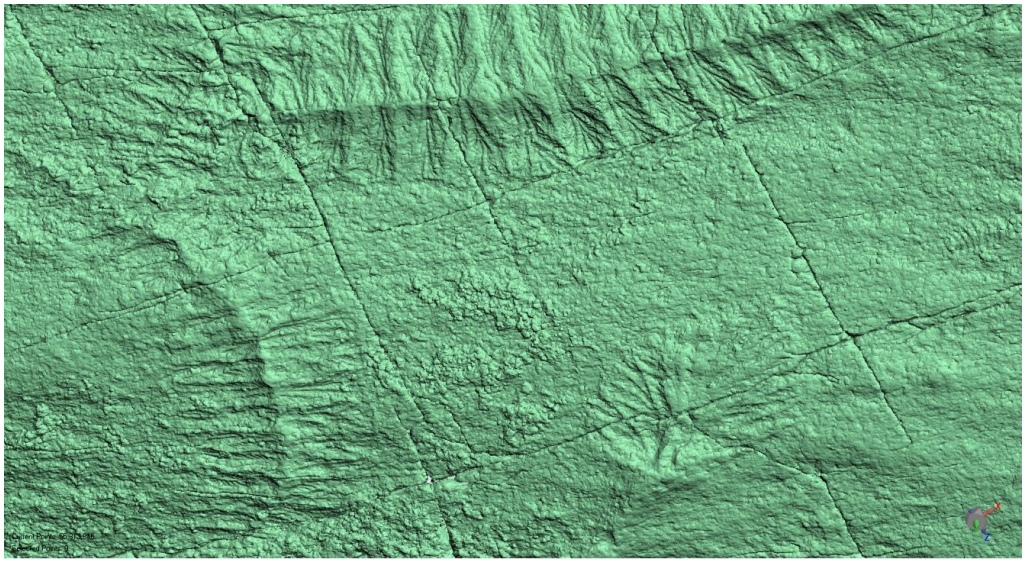
Laser scanning a room

WikiCommons

Light Detection And Ranging (**LiDAR**) uses laser scanning of external surfaces to create virtual 3D models which can be digitally manipulated

External Scanning: LiDAR example

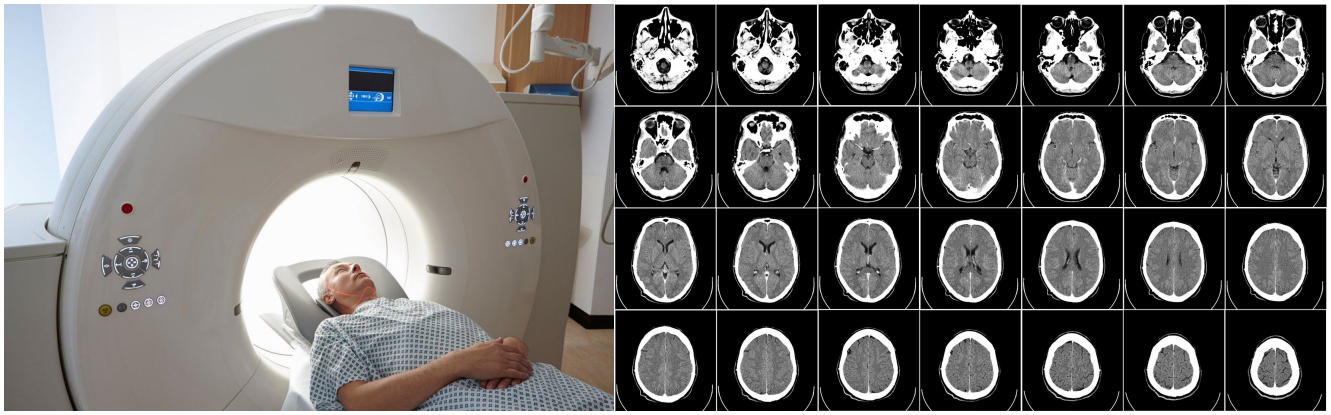




LiDAR being used to digitize and virtual preserve Ediacaran fossil localities in New Foundland

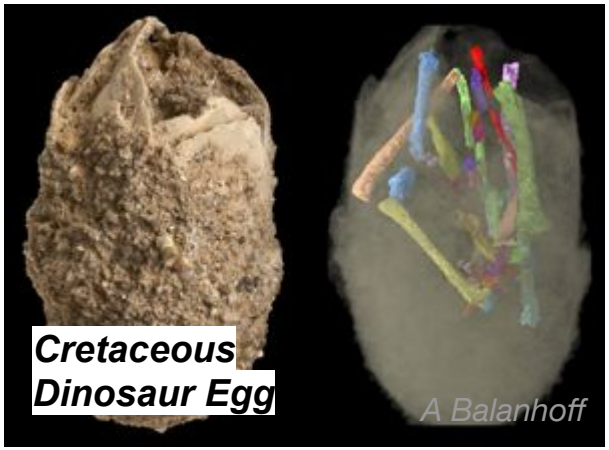
Ediacaran Example

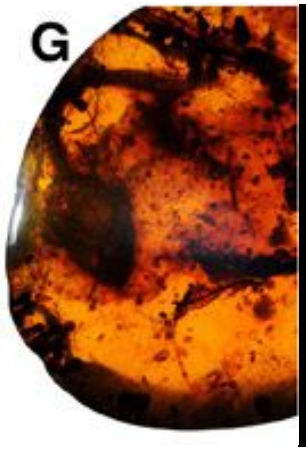
Internal Scanning: Computed Tomography (CT)

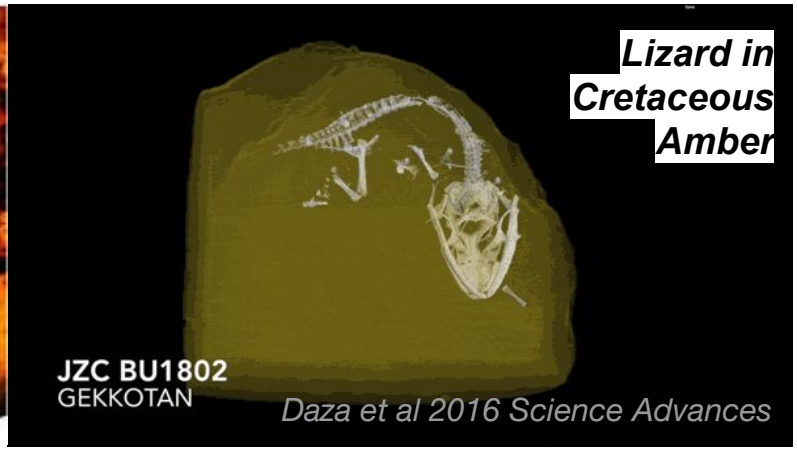


A series of X-ray scans

Non-invasive way to visualise internal anatomy

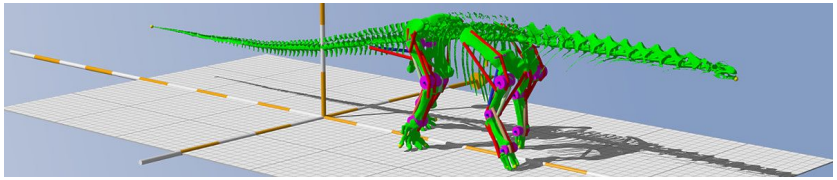




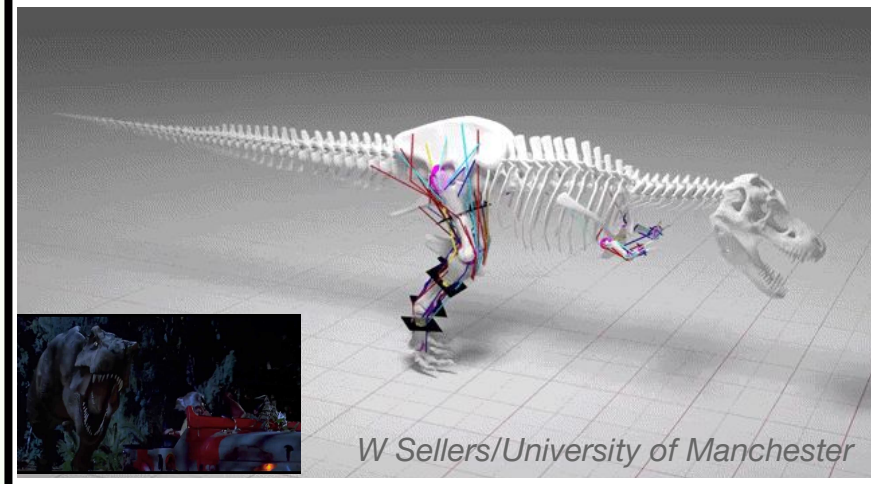


CT-scanning can visualize **internal anatomy** of fossils without destroying: images depends on attenuation contrast (e.g. absorption or scattering of X-rays)

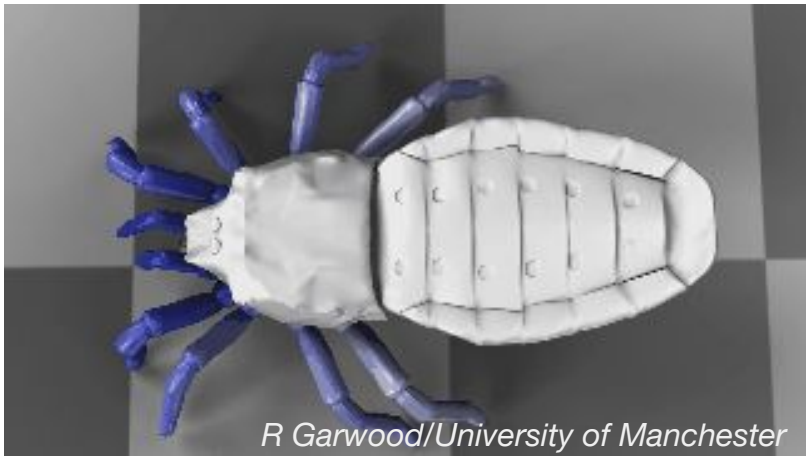
Biomechanics of Walking



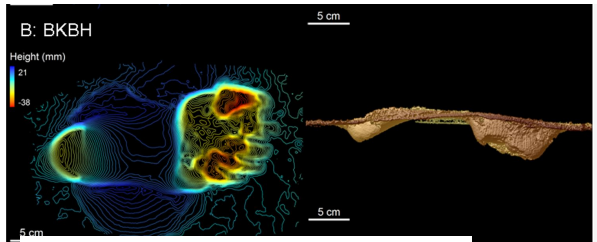
Digital models can be used to model walking e.g. applying virtual muscles and tendons to the skeleton



Gait reconstruction shows *T. rex* was unlikely to be able to run, and therefore not a hunter



Gait reconstruction of trigonotarbids show they walked a bit like spiders

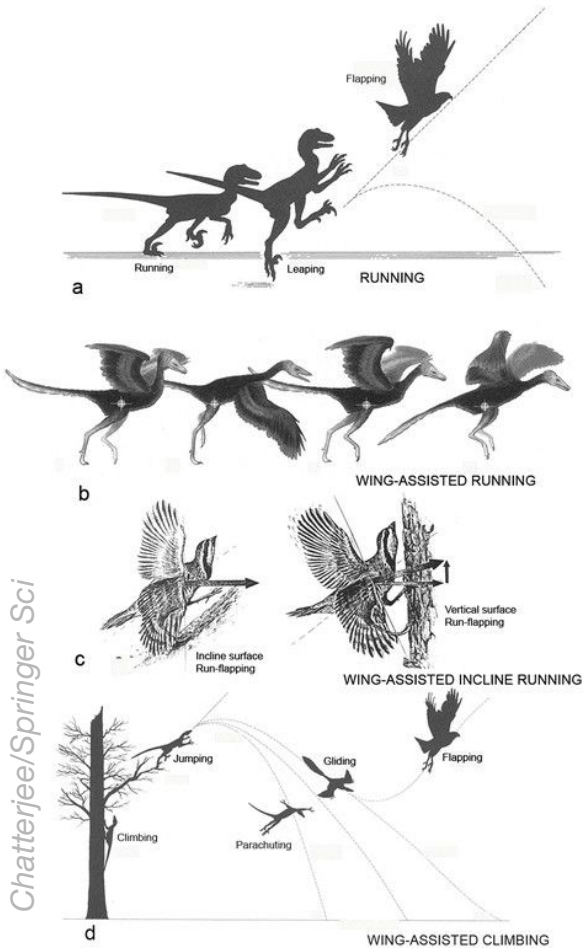


Laetoli foot prints 4.5mya

Raichlan et al 2010 PlosOne *Leakey Foundation*

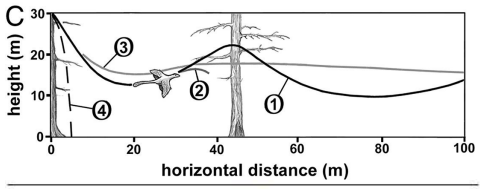
Foot prints and models demonstrate **bipedalism** in *Australopithecus*

Biomechanics of Flight



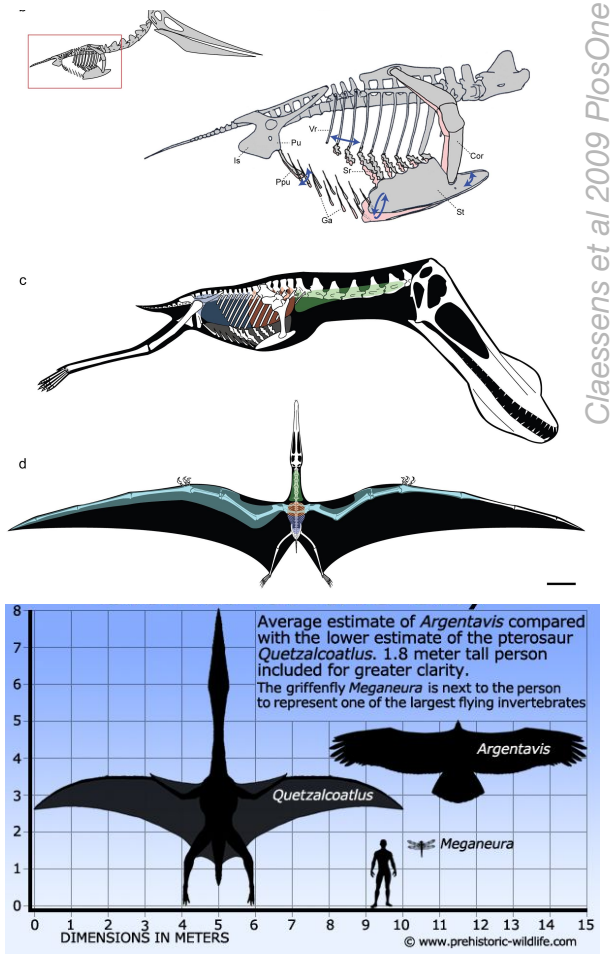
Chatterjee/Springer Sci

Different theories for the origin of bird flight



Chatterjee et al 2007 PNAS

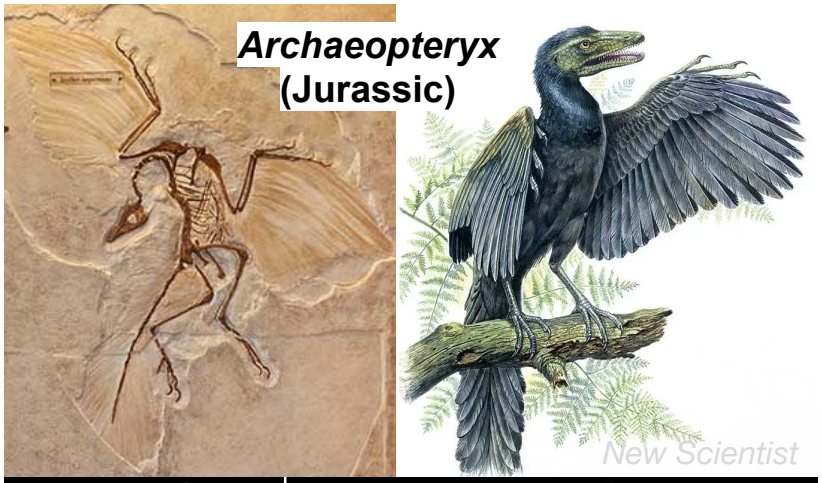
Reconstruction of *Microrapter* as glider

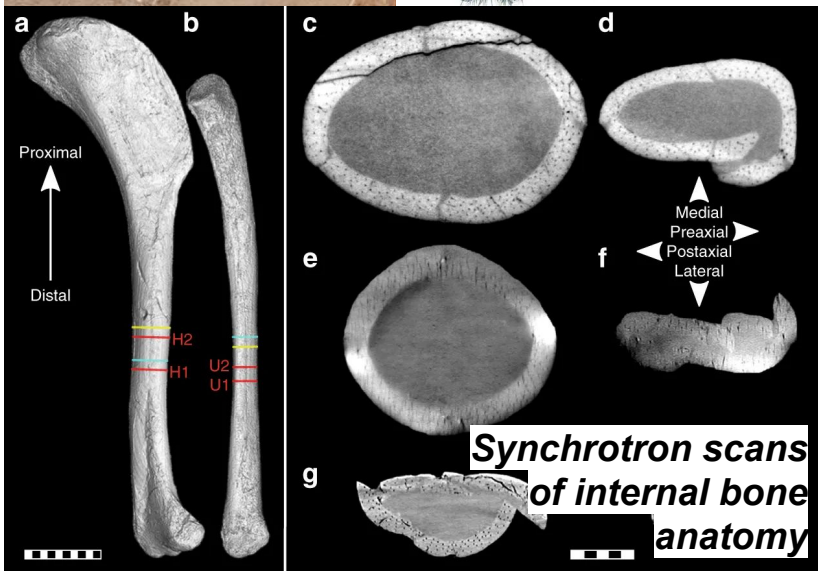


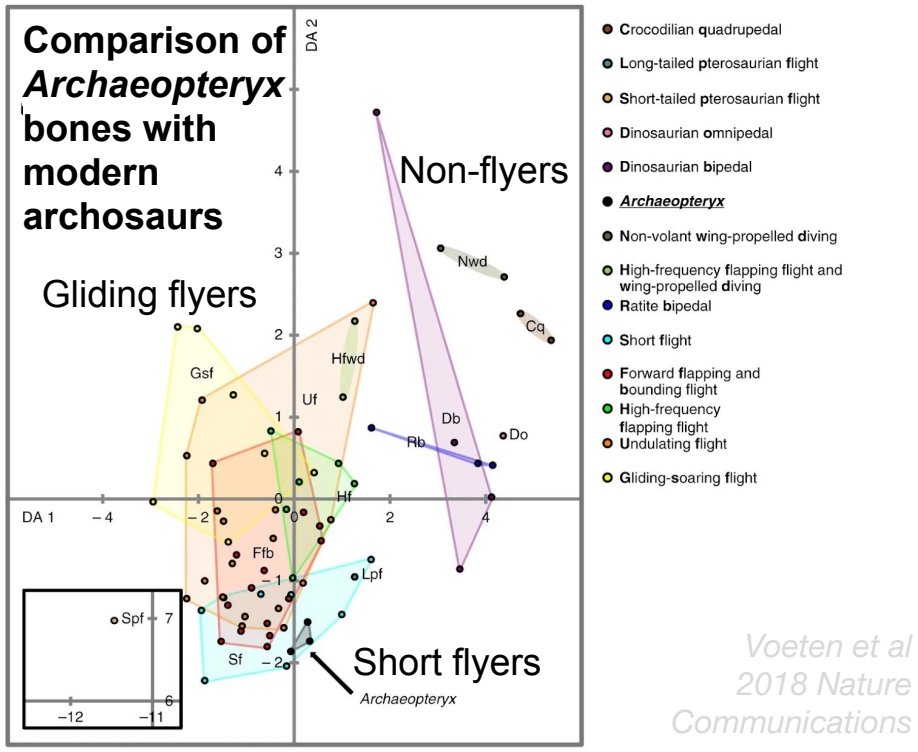
Claessens et al 2009 PlosOne

Pterosaurs had different flight, including giants

Biomechanics of Flight: an example



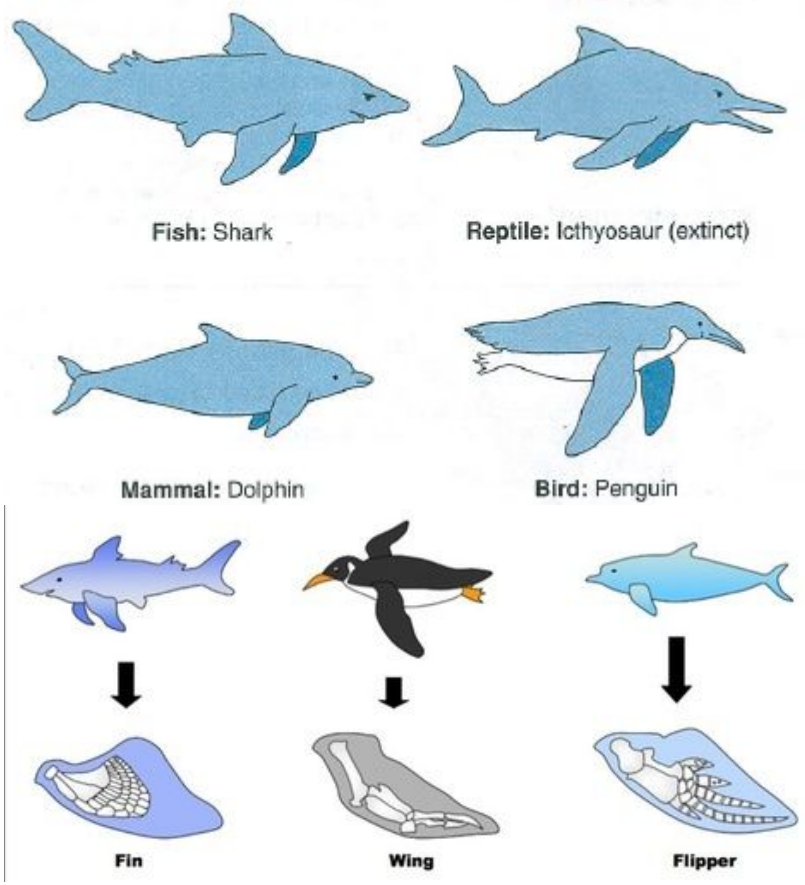




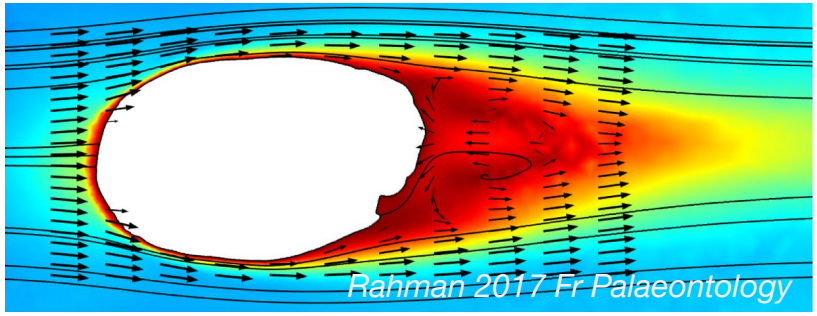
Voeten et al 2018 Nature Communications

The flight mode of *Archaeopteryx* reconstructed by comparing dimensions of bones with modern archosaurs *Archaeopteryx* compares closes with short burst flyers (e.g. pheasant)

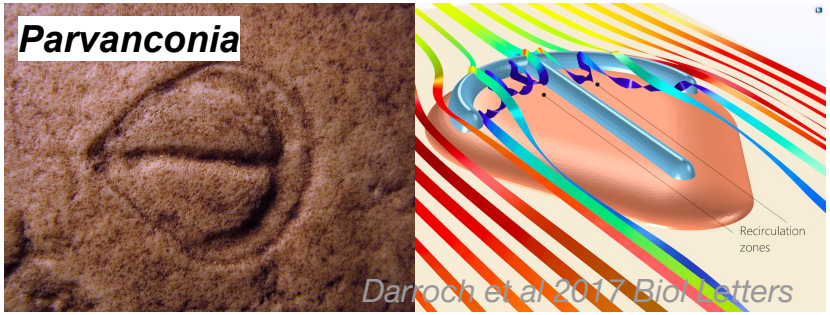
Biomechanics of Swimming



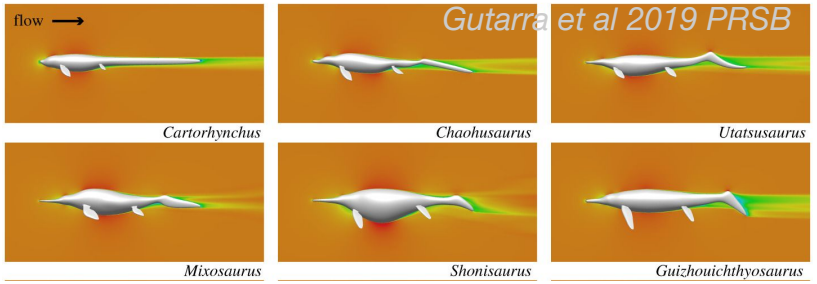
Fish, reptiles, mammals and birds have all **convergently** evolved the same **hydrodynamic** body shape



Computational fluid dynamics is an engineering technique to simulate flow

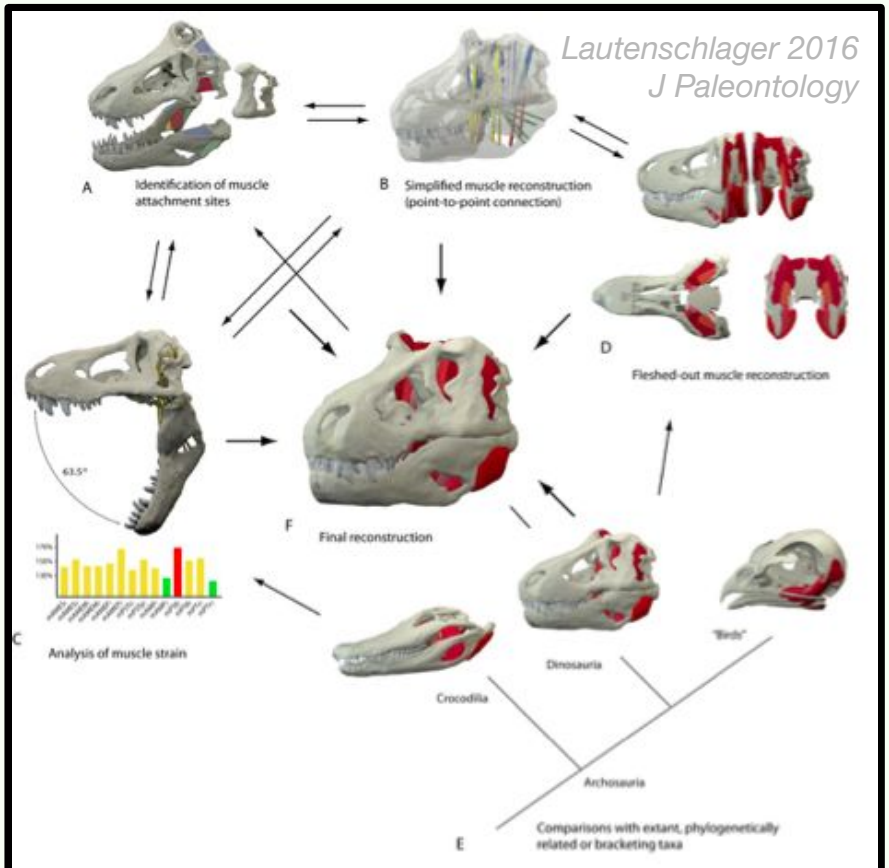


Ediacaran oddity likely suspension feeder, maybe orienting itself to flow

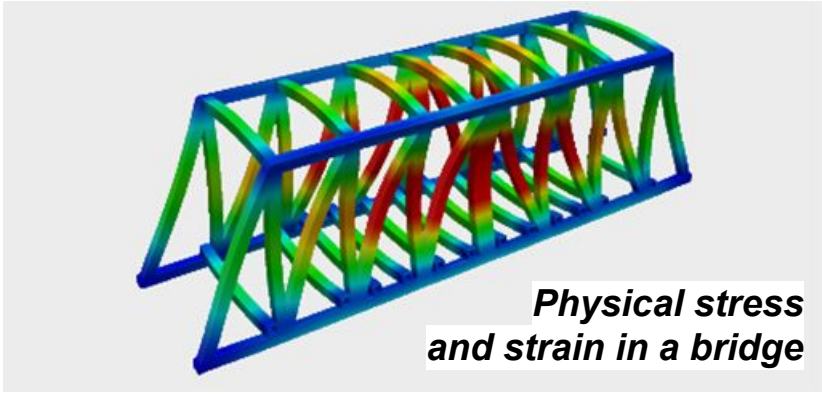


Increasingly large ichthyosaurs became relatively efficient swimmers

Biomechanics of Feeding

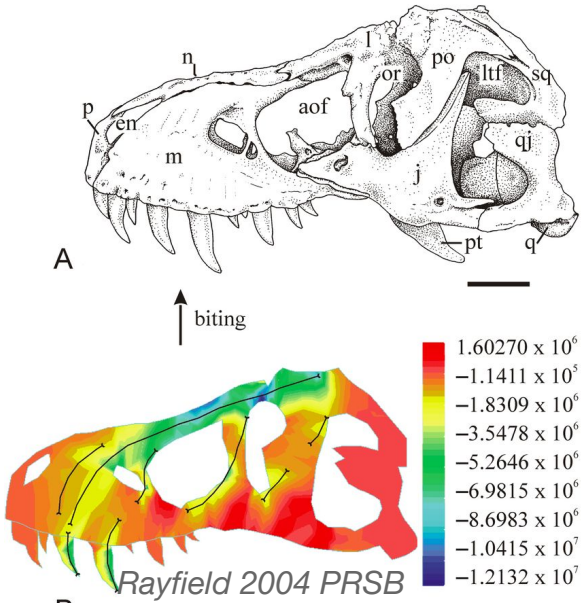


Soft-tissue **muscles** are reconstructed using **comparative anatomy**
Combined with osteology to infer biting strength and feeding styles



Physical stress and strain in a bridge

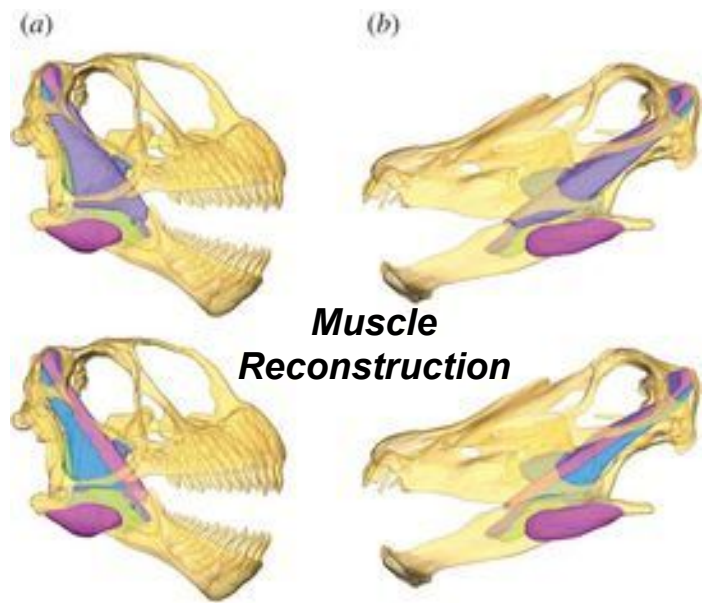
Finite Element Analysis (FEA) uses algebra to estimate physical properties of a simplified digital model



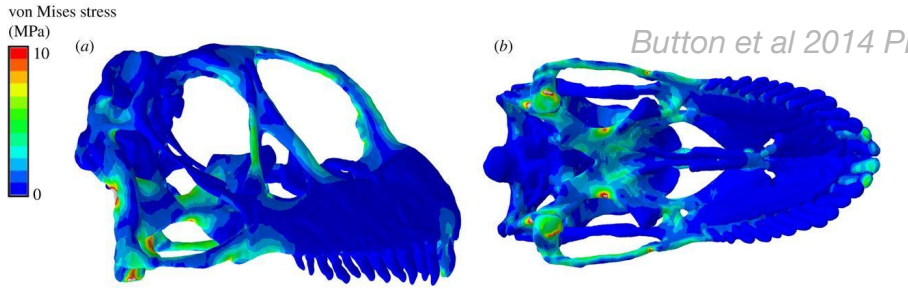
A *T. rex* model composed of smaller elements to model **stress** of biting and twisting in bone

"Puncture-Pull" interpretation of *T. rex* feeding supported

Biomechanics of Feeding: an example

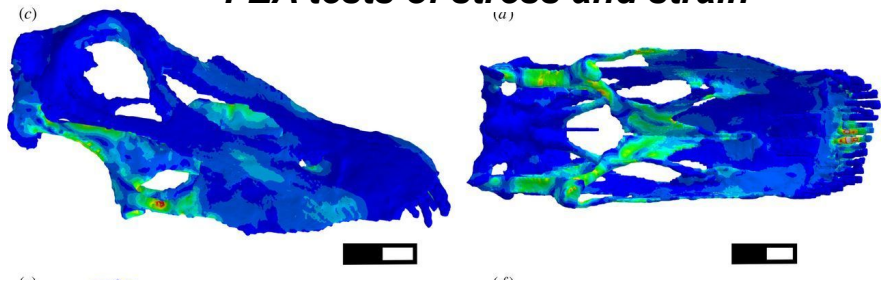


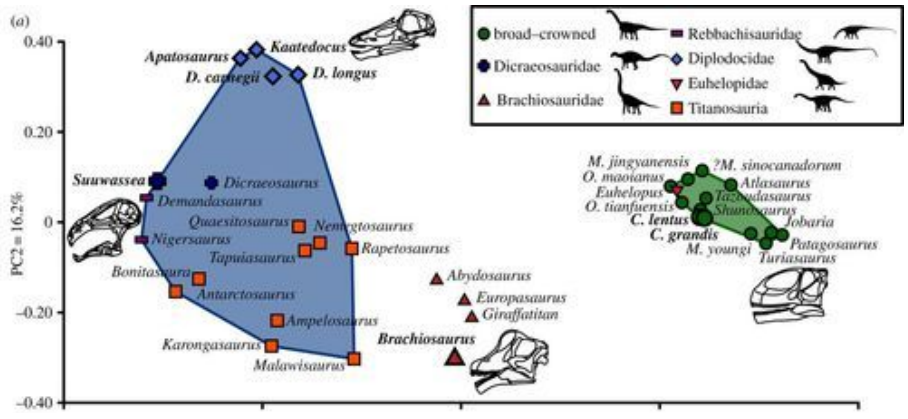
Muscle Reconstruction



Button et al 2014 PRSB

FEA tests of stress and strain





Reconstruction of feeding muscles, skull stress and strain in two sauropods (*Camarasaurus* & *Diplodocus*)

They had divergent specialization towards different food stuffs and strengths of bite forces

Dinosaurs!
7.4 The Real Jurassic Park: Engineering

Intended Learning Objective:
Describe, using examples, how engineering approaches are applied to fossil organisms to reconstruct anatomy, motion, behaviour

Now take a look at the associated task/question

Robert Sansom, U. Manchester

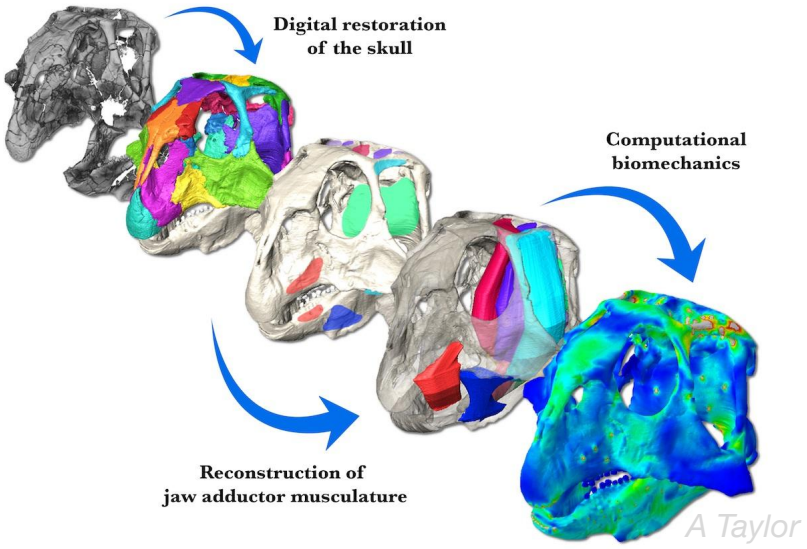
Part 7
Dinosaurs!

7.5 The Real Jurassic Park: Soft Tissues

EART11100: How to Grow a Planet
Robert Sansom
University of Manchester

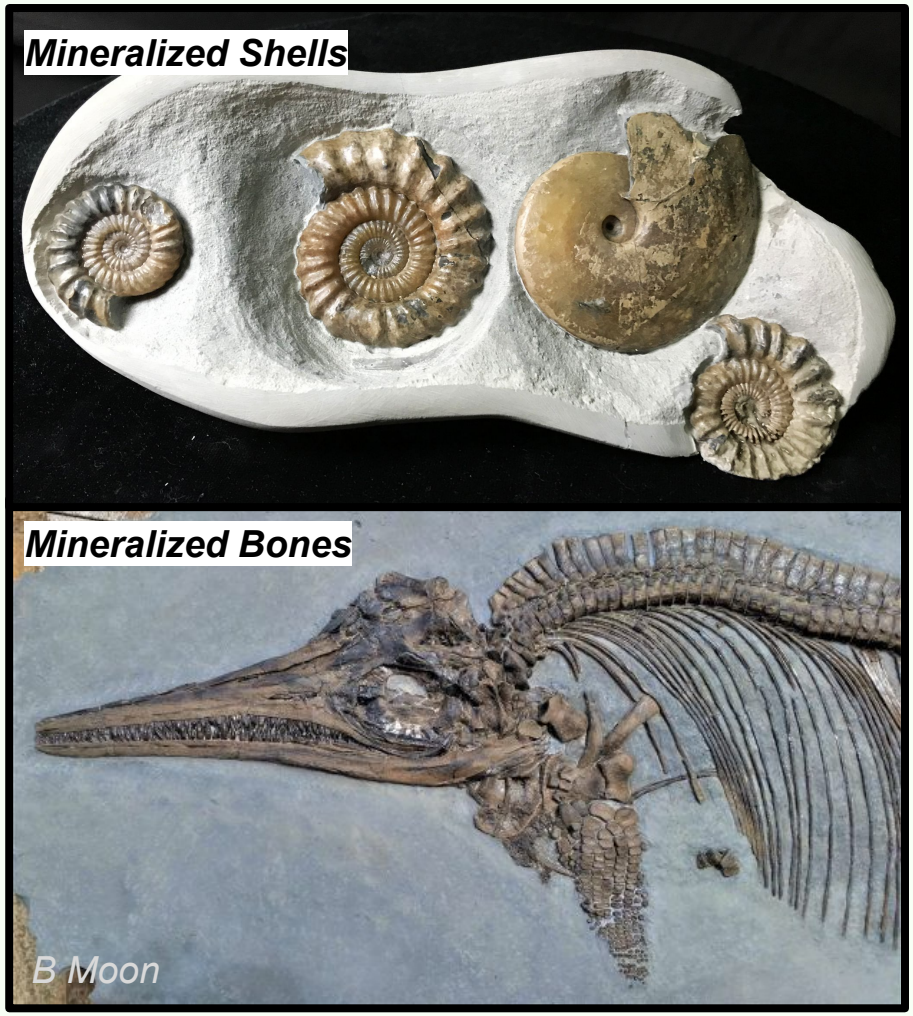
How can we bring the dead back to life?

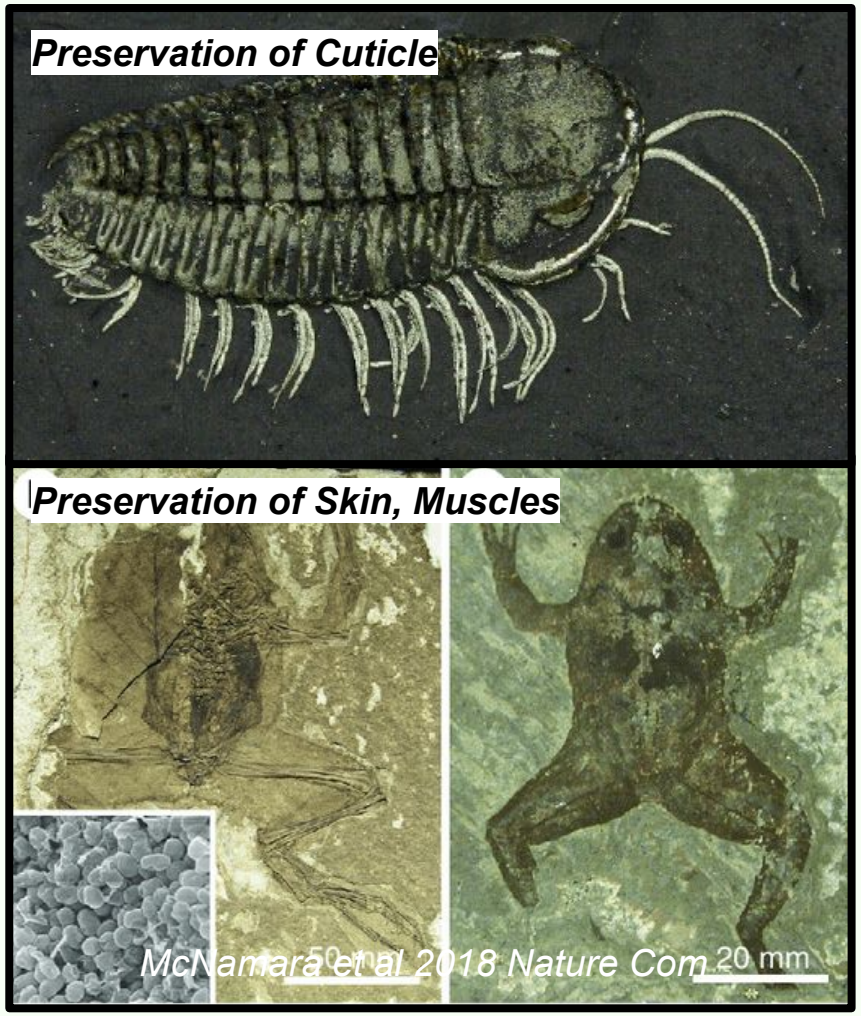
Engineering approaches can help us reconstruct the motion and behaviour of long extinct animals, for example feeding and locomotion



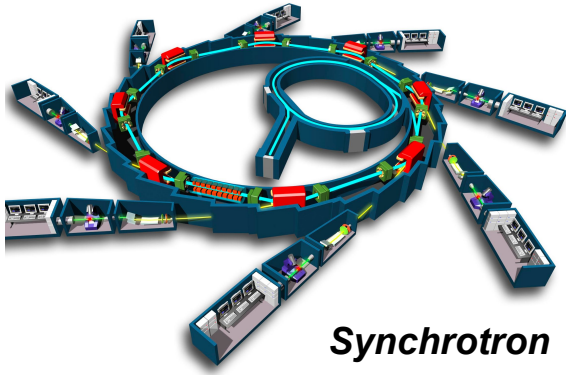
Studying exceptionally preserved **soft tissues** can provide insights into appearance and behaviour of extinct animals, for example colours

Hard Tissues and Soft Tissues

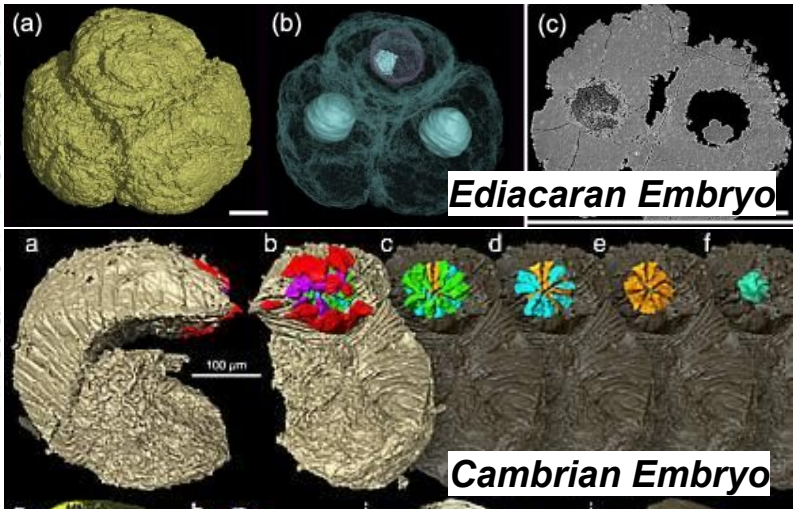




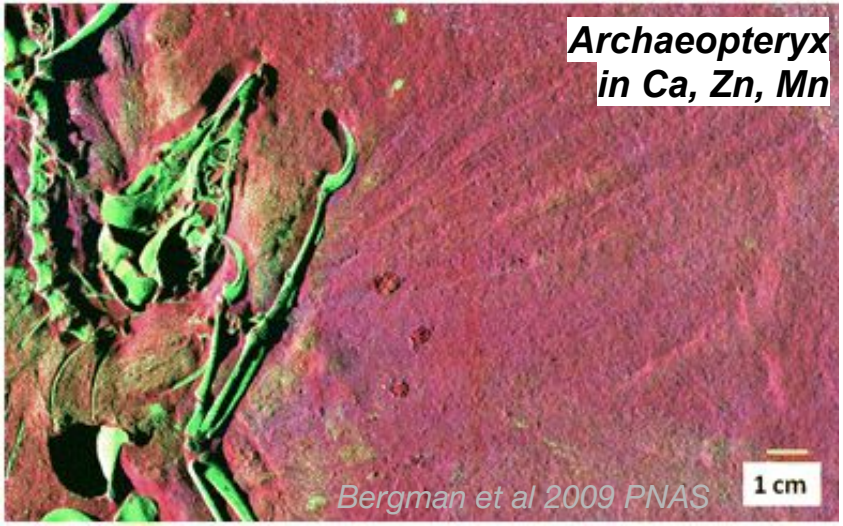
Scanning Soft Tissues



A **synchrotron** is a type of particle accelerator that generates high power X-rays
Many applications including palaeobiology

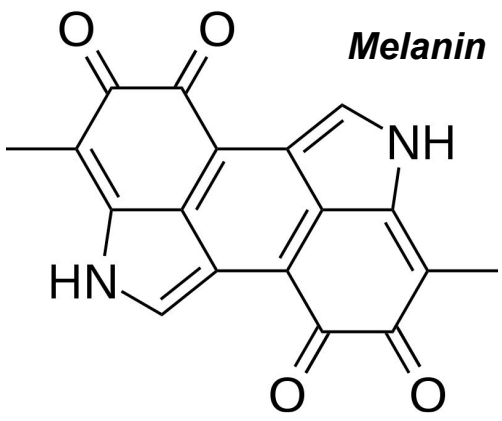


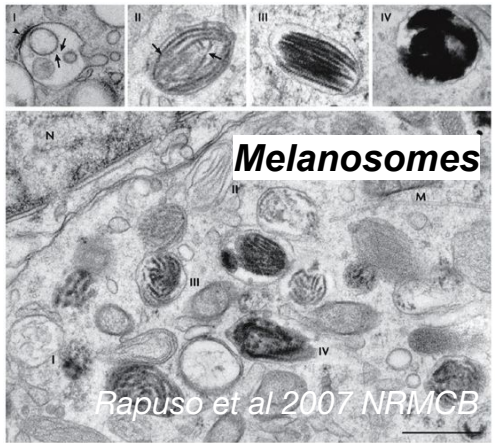
Synchrotron tomography of tiny fossil embryos reveals Ediacaran nuclei & Cambrian animal blastulae



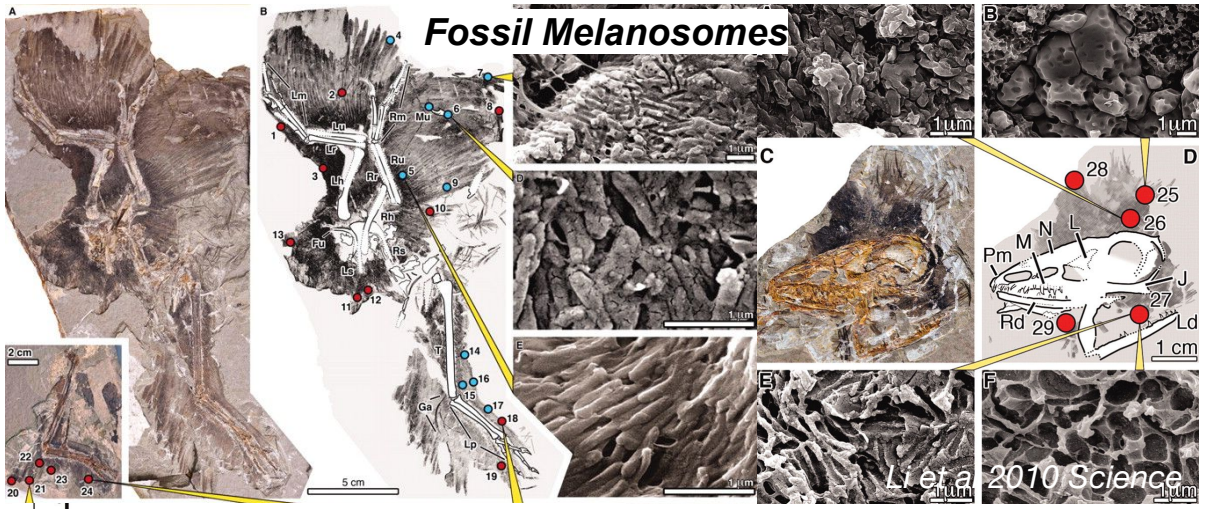
X-ray fluorescence (XRF) allows detection of the chemical elements and their distribution in detail

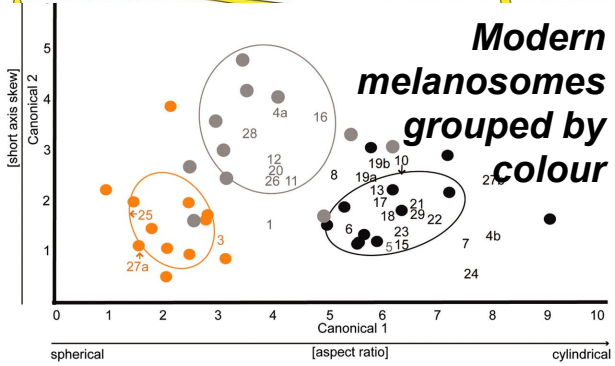
Fossil Colours: Melanosomes

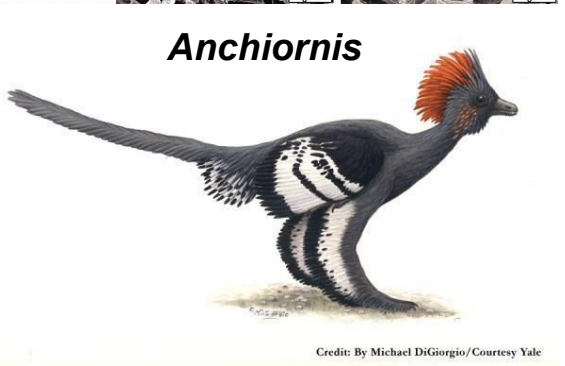




Melanin is a natural pigment packaged in sub-cellular **melanosomes**



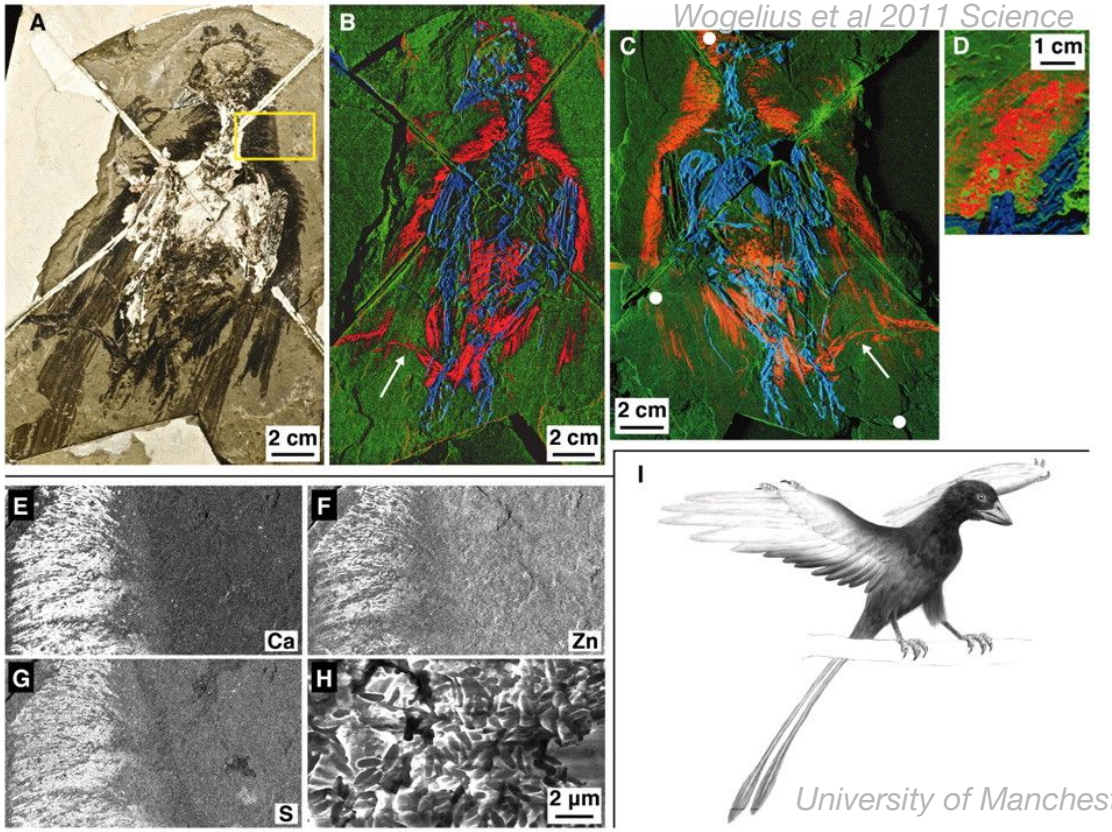




Different coloured melanosomes have different shapes and densities
They can be preserved in fossils which allows reconstruction of distribution of colour pigments in life

Fossil Colour: Metals

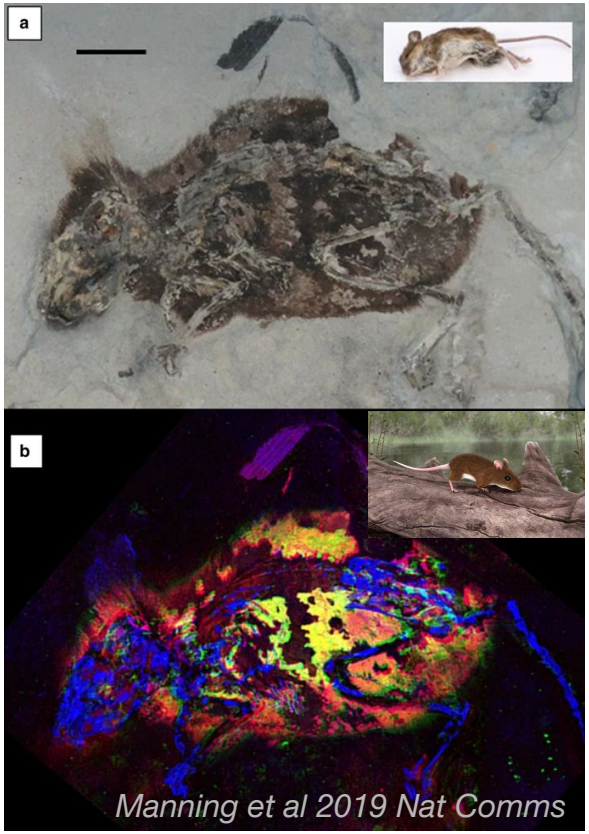
Synchrotron XRF can also be used to detect trace elements related to pigment



Wogelius et al 2011 Science

University of Manchester

Scanning a fossil bird finds copper likely derived from eumelanin, thus reconstruction colour patterns

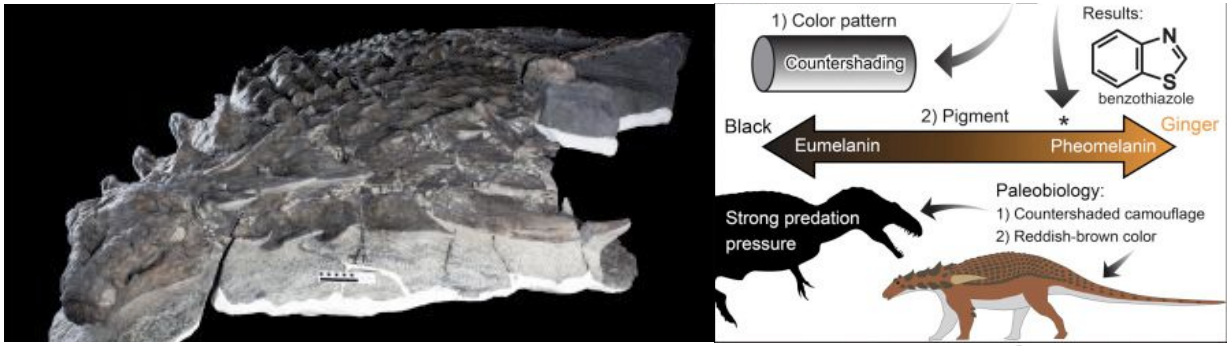


Manning et al 2019 Nat Comms

Fossil mouse with pheomelanin related Zinc

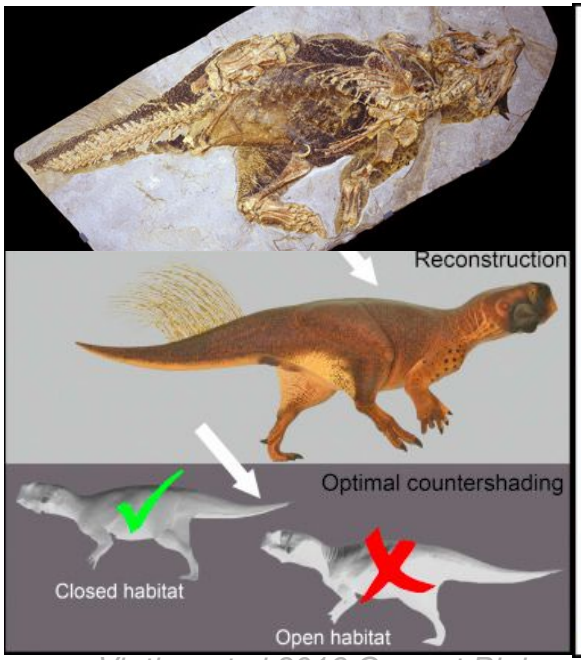
Fossil Colours: Ecology

The distribution of colour can be used to infer animal ecology and behaviour

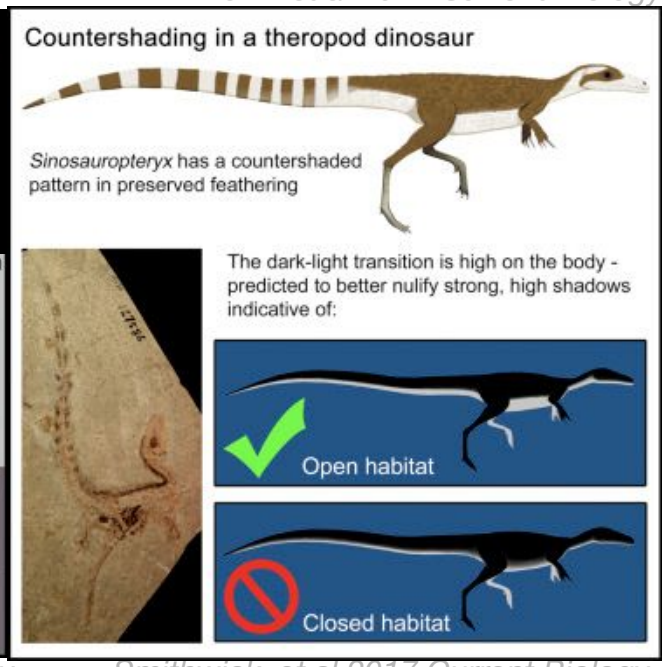


Brown et al 2017 Current Biology

These dinosaurs exhibited counter-shading, probably to evade predation



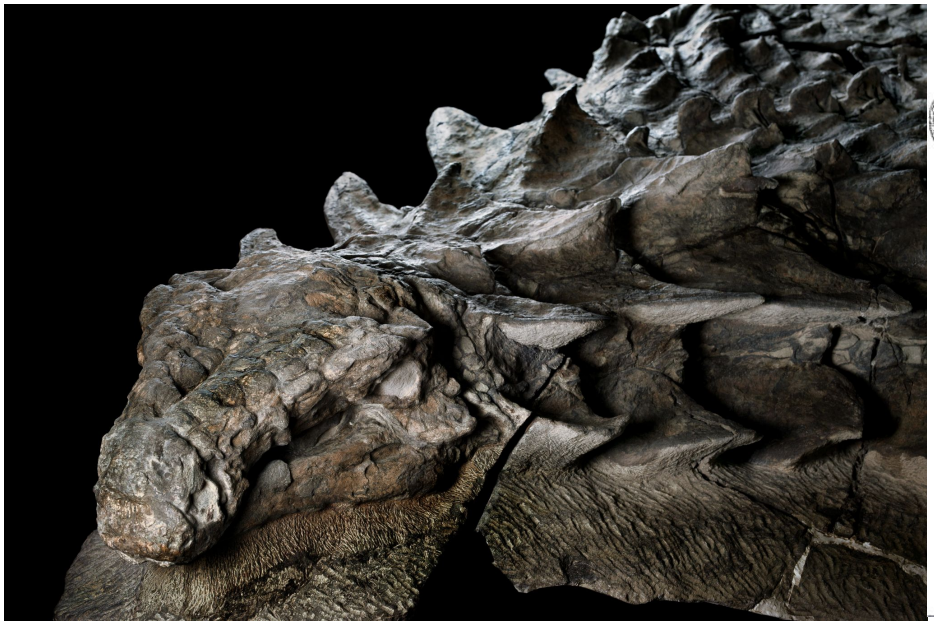
Vinther et al 2016 Current Biology

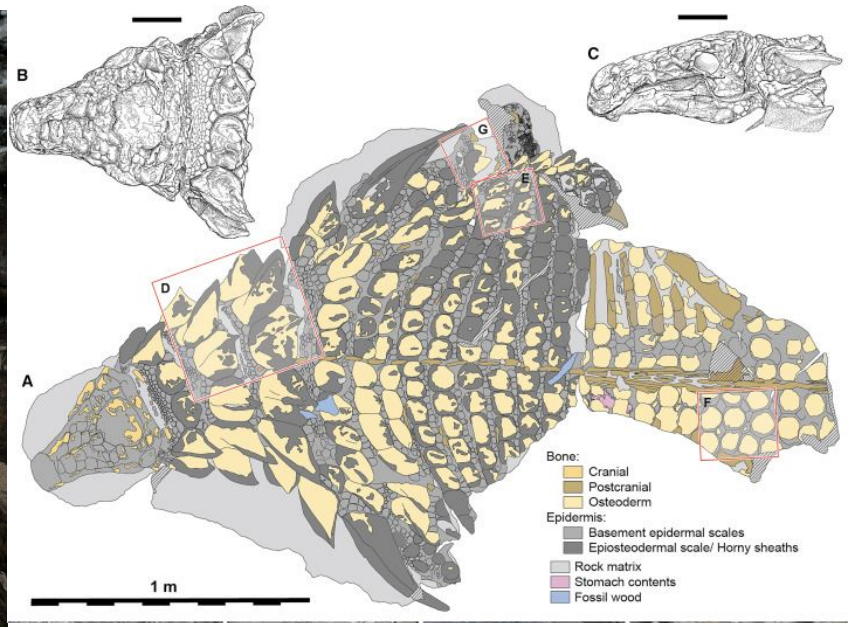


Smithwick et al 2017 Current Biology

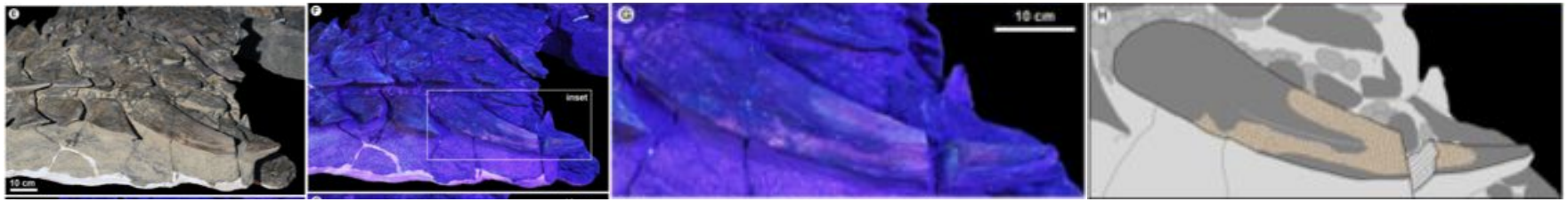
The patterns can also infer habitat type as well as behaviour

Fossil Colours: Ecology





Borealopelta is preserved in really fine detail including osteoderms and armour



Brown et al 2017 Current Biology

Dinosaurs!
7.5 The Real Jurassic Park: Soft Tissues

Intended Learning Objective: Describe how modern scanning technologies can shed light on the internal anatomy and soft tissues of fossil organisms and thus the affinity and behaviour of extinct organisms

Robert Sansom, University of Manchester

Universal Studios

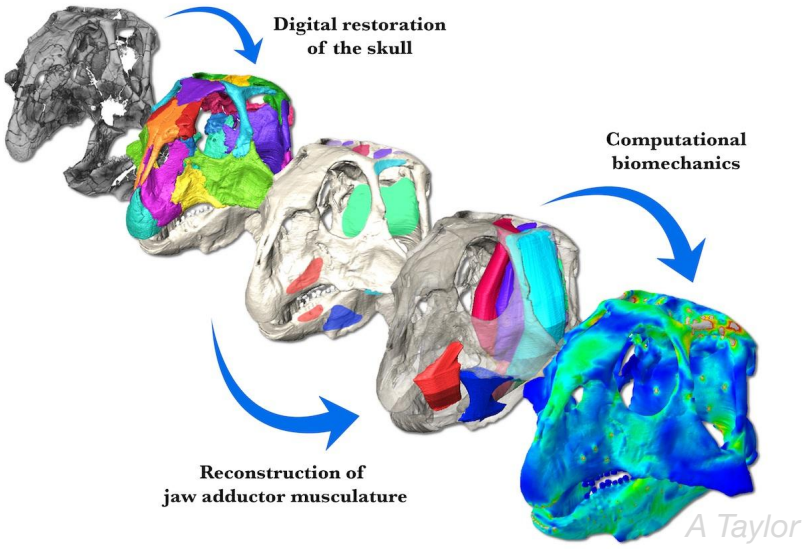
Part 7
Dinosaurs!

7.6 The Real Jurassic Park: Genetics

EART11100: How to Grow a Planet
Robert Sansom
University of Manchester

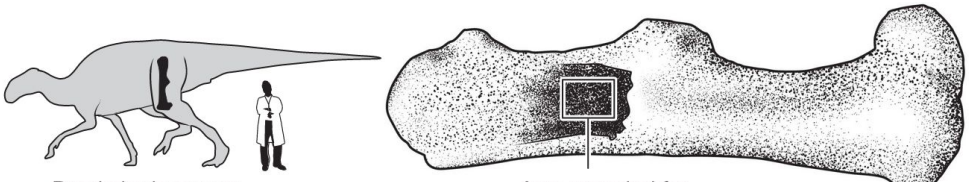
How can we bring the dead back to life?

Engineering approaches can help us reconstruct the motion and behaviour of long extinct animals, for example feeding and locomotion



Studying exceptionally preserved **soft tissues** can provide insights into appearance and behaviour of extinct animals, for example colours

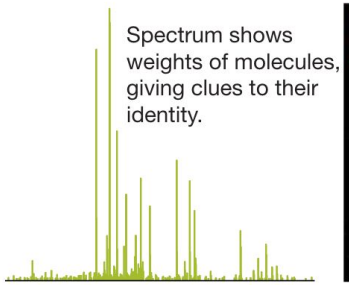
Other Soft Tissues from Dinosaurs?



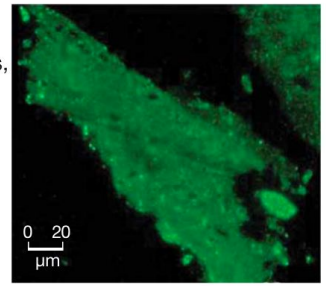
Brachylophosaurus canadensis femur

Area sampled for molecular analysis

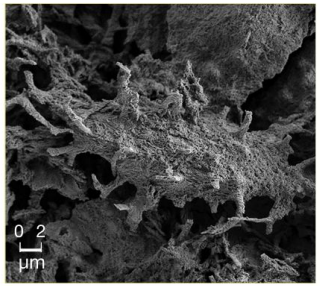
Controversial clues



Mass spectrometry
Identifies protein fragments and provides the sequence of amino acids that make up each fragment.



Antibody fluorescence
Lights up target proteins when fluorescent-labeled antibodies bind to them.



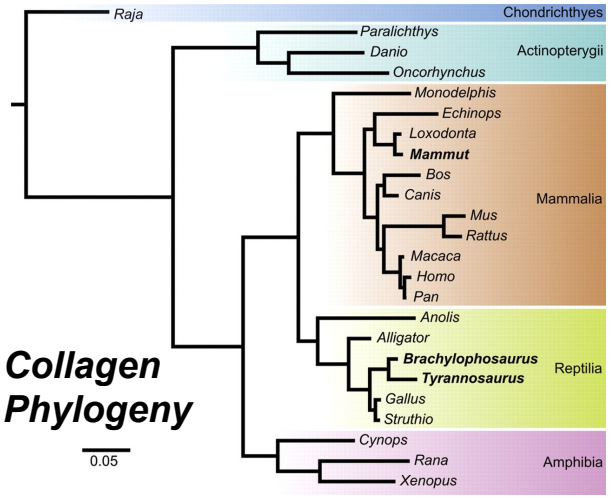
Microscopy
Reveals features in demineralized fossils. Here, what appears to be a bone cell nestles within a fibrous matrix that may be collagen.

One USA research group claim to have extracted proteins and cells from dinosaur bone tissue, including *T. rex*
However, nobody else has been able to replicate these results.

Schweitzer et al 2009 Science

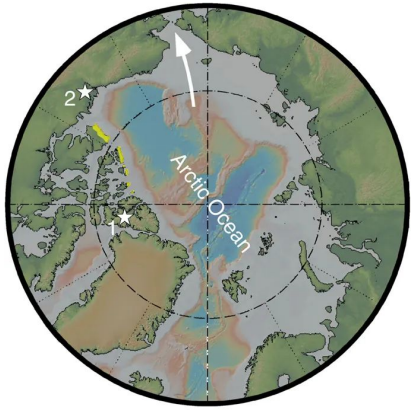
```
                                                  GATGAPGIAGAPGFPG
ARGAPGPQGPSGAPGPK----------------------------------------------------
----------------------------------GVQGPPGPQGP----------------------
----------------------------------RGSAGPPGATGFPGAAGR----------------
-----GVVGLPGQR-------------------------------------------------------
```

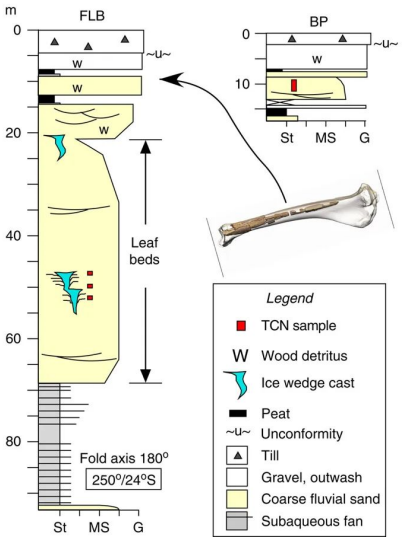
Tyrannosaurus collagen protein sequence



Collagen Phylogeny

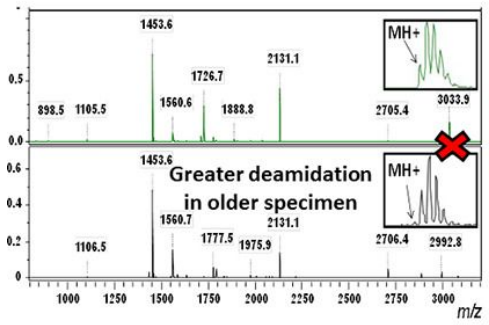
Fossil proteins



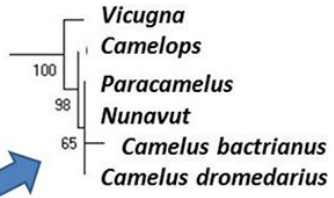


Bone fragment from Arctic Canada, unknown affinity

Rybczynski et al Nat Com 2013
Buckley et al 2019 J Proteomics



Diagenetic modifications result in few missing peptides

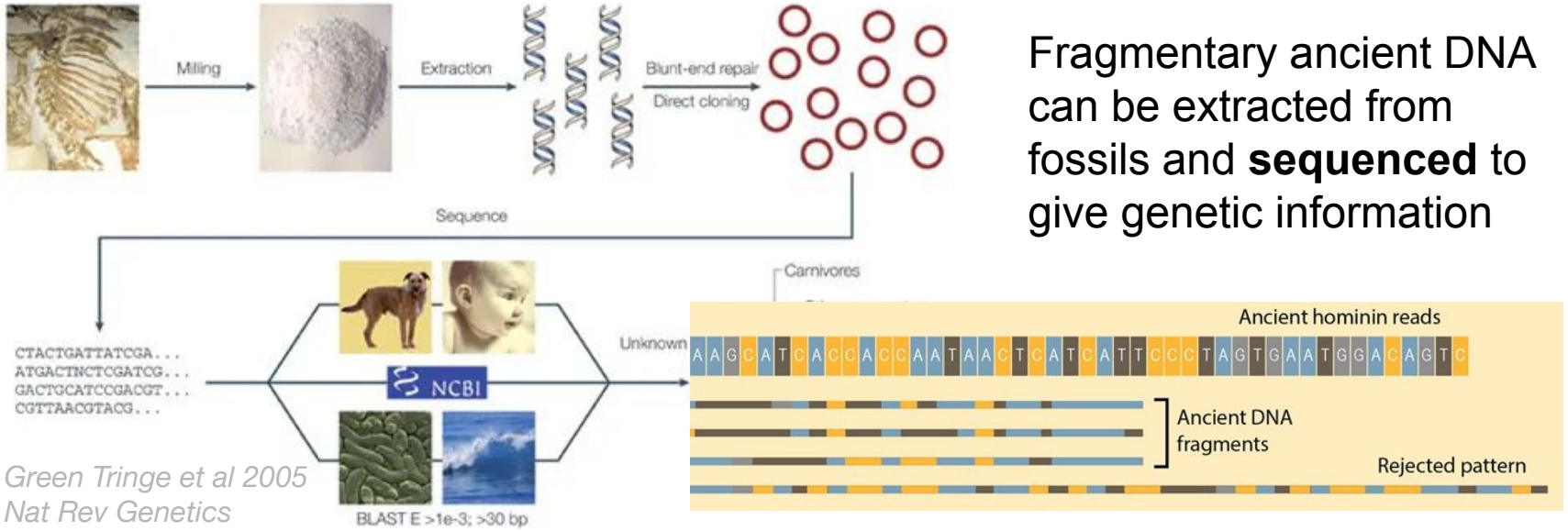


Omission of a single peptide results in phylogenetic topology change

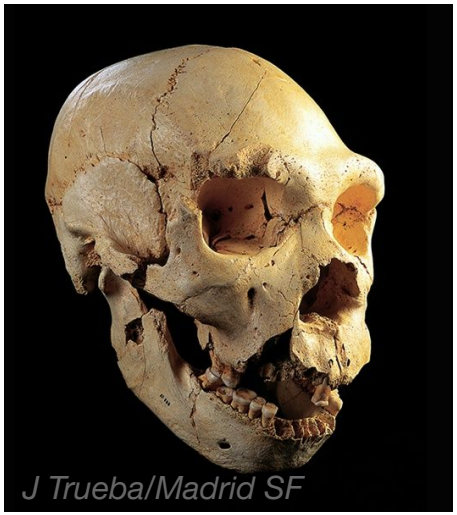
Collagen proteins extracted and sequenced: revealed bone to be from a Camel

Camels migrated to North America via Arctic during warm period, 3.5 million years ago

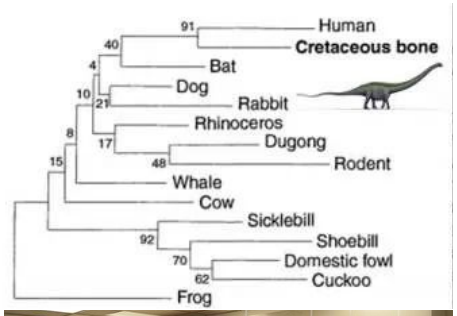
Ancient DNA



Fragmentary ancient DNA can be extracted from fossils and **sequenced** to give genetic information



DNA sequences extracted from Neanderthals reveal extensive interbreeding between neanderthals & *Homo sapiens*



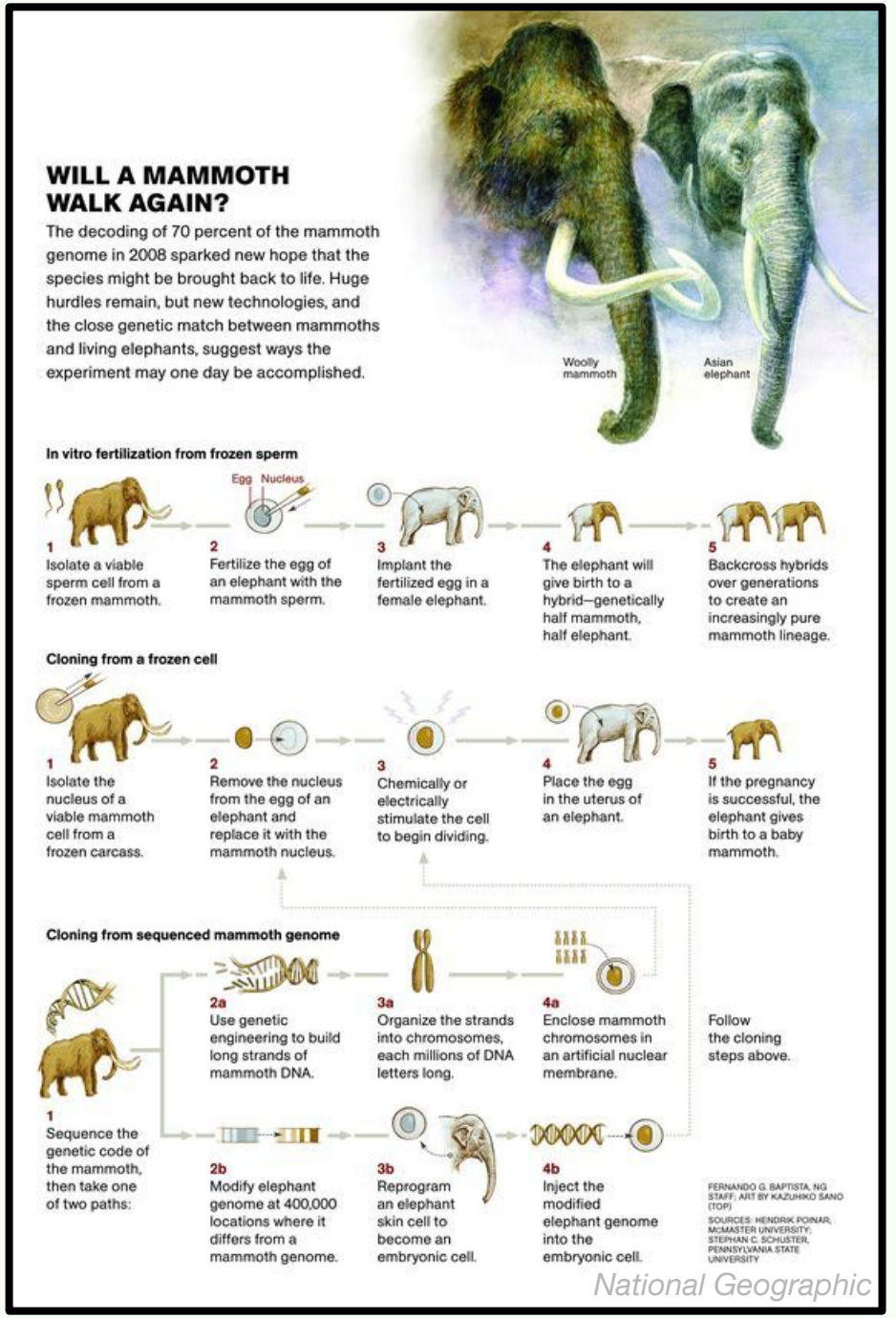
Highly sensitive to contamination and highly labile

Oldest known reliable ancient DNA is 800,000 years old

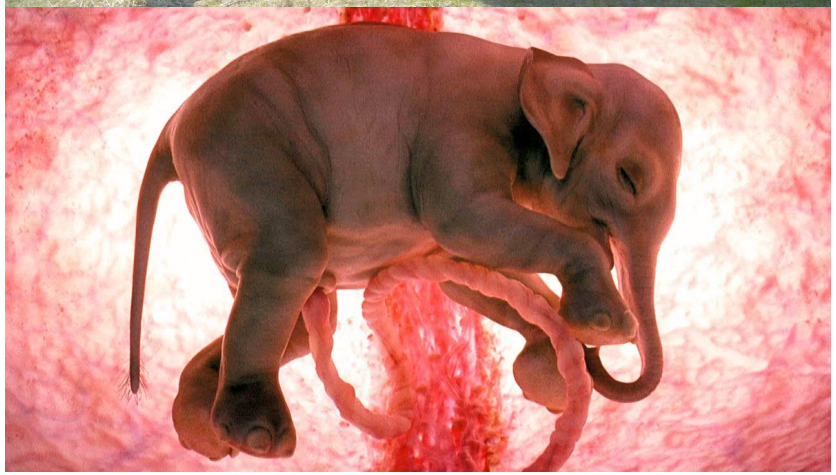
Clone a Mammoth?

Could the wooly mammoth be brought back from extinction through **dextinction**?

Possible routes include cloning from frozen cells or sequenced genome.



Ethics

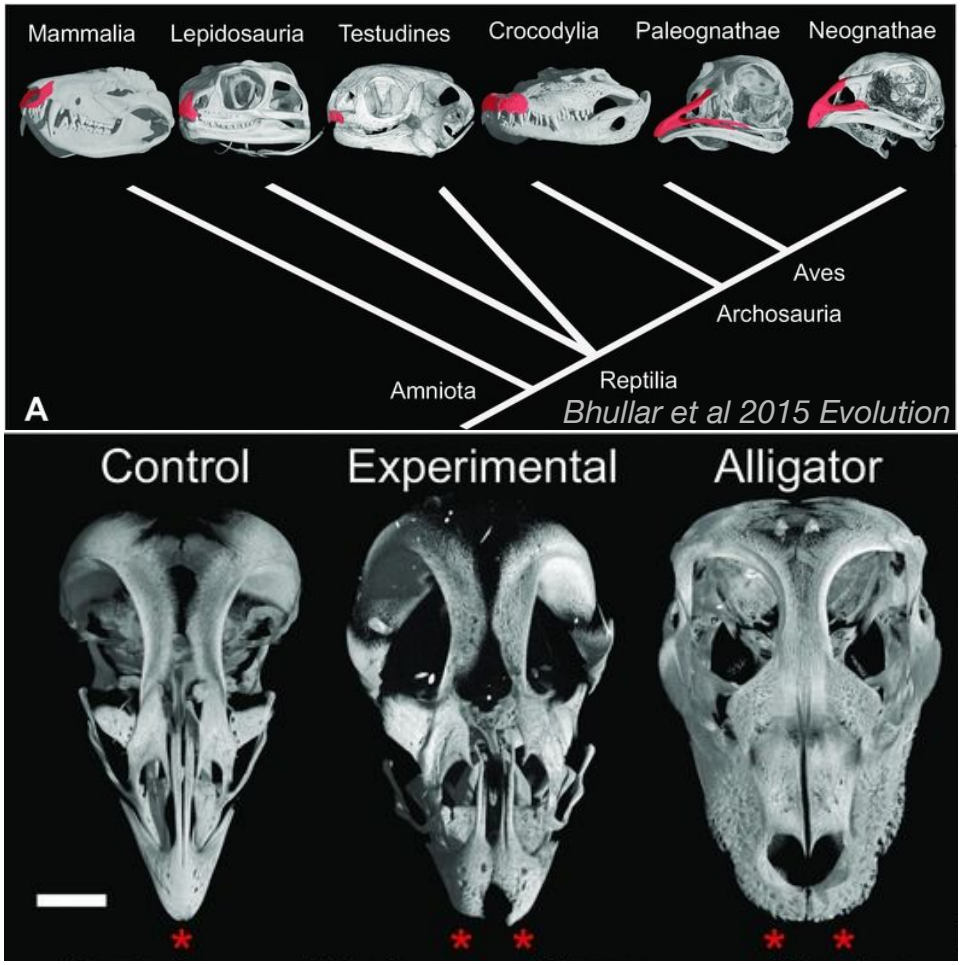


Should we bring back mammoths?
Do we have a responsibility to?

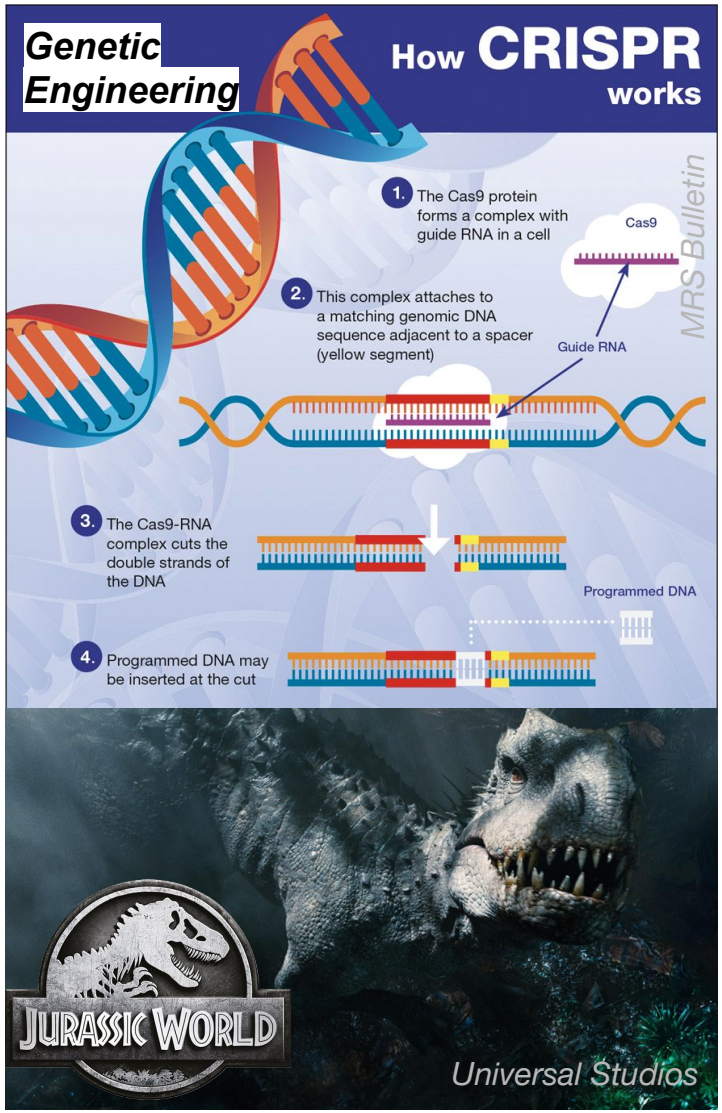
Ecosystem Engineers?

Hunting?

Dino-Chicken?



Disrupting the development of the beak in a chicken gives a dinosaur-type skull



Dinosaurs!
7.6 The Real Jurassic Park: Genetics

Intended Learning Objective: Evaluate the limitations and ethical implications of genetic engineering and cloning approaches to de-extinction

Now take a look at the associated task/question

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